

Part IB — Methods

Based on lectures by D. B. Skinner

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These notes are not endorsed by the lecturers, and I have modified them (often significantly) after lectures. They are nowhere near accurate representations of what was actually lectured, and in particular, all errors are almost surely mine.

Self-adjoint ODEs

Periodic functions. Fourier series: definition and simple properties; Parseval's theorem. Equations of second order. Self-adjoint differential operators. The Sturm-Liouville equation; eigenfunctions and eigenvalues; reality of eigenvalues and orthogonality of eigenfunctions; eigenfunction expansions (Fourier series as prototype), approximation in mean square, statement of completeness. [5]

PDEs on bounded domains: separation of variables

Physical basis of Laplace's equation, the wave equation and the diffusion equation. General method of separation of variables in Cartesian, cylindrical and spherical coordinates. Legendre's equation: derivation, solutions including explicit forms of P_0 , P_1 and P_2 , orthogonality. Bessel's equation of integer order as an example of a self-adjoint eigenvalue problem with non-trivial weight.

Examples including potentials on rectangular and circular domains and on a spherical domain (axisymmetric case only), waves on a finite string and heat flow down a semi-infinite rod. [5]

Inhomogeneous ODEs: Green's functions

Properties of the Dirac delta function. Initial value problems and forced problems with two fixed end points; solution using Green's functions. Eigenfunction expansions of the delta function and Green's functions. [4]

Fourier transforms

Fourier transforms: definition and simple properties; inversion and convolution theorems. The discrete Fourier transform. Examples of application to linear systems. Relationship of transfer function to Green's function for initial value problems. [4]

PDEs on unbounded domains

Classification of PDEs in two independent variables. Well posedness. Solution by the method of characteristics. Green's functions for PDEs in 1, 2 and 3 independent variables; fundamental solutions of the wave equation, Laplace's equation and the diffusion equation. The method of images. Application to the forced wave equation, Poisson's equation and forced diffusion equation. Transient solutions of diffusion problems: the error function. [6]

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0 Introduction

In the previous courses, the (partial) differential equations we have seen are mostly linear. For example, we have Laplace's equation:

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0,$$

and the heat equation:

$$\frac{\partial \phi}{\partial t} = \kappa \left(\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} \right).$$

The Schrödinger' equation in quantum mechanics is also linear:

$$i\hbar \frac{\partial \Phi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Phi}{\partial x^2} + V(x)\Phi(x).$$

By being linear, these equations have the property that if ϕ_1, ϕ_2 are solutions, then so are $\lambda_1 \phi_1 + \lambda_2 \phi_2$ (for any constants λ_i).

Why are all these linear? In general, if we just randomly write down a differential equation, most likely it is not going to be linear. So where did the linearity of equations of physics come from?

The answer is that the real world is *not* linear in general. However, often we are not looking for a completely accurate and precise description of the universe. When we have low energy/speed/whatever, we can often quite accurately approximate reality by a linear equation. For example, the equation of general relativity is very complicated and nowhere near being linear, but for small masses and velocities, they reduce to Newton's law of gravitation, which is linear.

The only exception to this seems to be Schrödinger's equation. While there are many theories and equations that superseded the Schrödinger equation, these are all still linear in nature. It seems that linearity is the thing that underpins quantum mechanics.

Due to the prevalence of linear equations, it is rather important that we understand these equations well, and this is our primary objective of the course.

1 Vector spaces

In studying differential equations, we frequently encounter spaces of functions that possess a natural vector space structure. Given a set of basis functions $\{y_n\}$, we seek to expand an arbitrary function f as

$$f(x) = \sum_n c_n y_n(x)$$

for some coefficients c_n . A familiar example is the Taylor series expansion

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n,$$

where the monomials $\{x^n : n \in \mathbb{N}\}$ serve as basis functions. The choice of basis depends on the problem at hand: for periodic functions, we will use periodic basis elements (Fourier series); for other problems, different bases prove more natural.

Two fundamental questions arise in this context:

- (i) Given a basis $\{y_n\}$, how do we compute the expansion coefficients c_n ?
- (ii) When does the series $\sum_n c_n y_n(x)$ converge, and does it converge to f ?

Linear algebra provides the framework for addressing the first question. Although classical linear algebra deals with finite-dimensional spaces, the techniques extend to infinite-dimensional function spaces under appropriate conditions.

We begin with the basic definitions.

Definition (Vector space). A *vector space* over $\mathbb{C}$ (or $\mathbb{R}$) is a set V equipped with an addition operation $+: V \times V \rightarrow V$ and a scalar multiplication $\cdot: \mathbb{C} \times V \rightarrow V$ satisfying the following axioms for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and all $\lambda, \mu \in \mathbb{C}$:

- (i) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ (commutativity)
- (ii) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ (associativity of addition)
- (iii) There exists $\mathbf{0} \in V$ such that $\mathbf{0} + \mathbf{u} = \mathbf{u}$ (additive identity)
- (iv) For each $\mathbf{u} \in V$, there exists $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ (additive inverse)
- (v) $\lambda(\mu\mathbf{v}) = (\lambda\mu)\mathbf{v}$ (associativity of scalar multiplication)
- (vi) $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}$ (distributivity over vector addition)
- (vii) $(\lambda + \mu)\mathbf{u} = \lambda\mathbf{u} + \mu\mathbf{u}$ (distributivity over scalar addition)
- (viii) $1\mathbf{v} = \mathbf{v}$ (multiplicative identity)

A vector space equipped with additional structure becomes more useful for analysis. The most important such structure is an inner product.

Definition (Inner product). An *inner product* on a vector space V over $\mathbb{C}$ is a map $(\cdot, \cdot) : V \times V \rightarrow \mathbb{C}$ satisfying, for all $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and $\lambda \in \mathbb{C}$:

- (i) $(\mathbf{u}, \lambda \mathbf{v}) = \lambda(\mathbf{u}, \mathbf{v})$ (linearity in second argument)
- (ii) $(\mathbf{u}, \mathbf{v} + \mathbf{w}) = (\mathbf{u}, \mathbf{v}) + (\mathbf{u}, \mathbf{w})$ (additivity in second argument)
- (iii) $(\mathbf{u}, \mathbf{v}) = (\mathbf{v}, \mathbf{u})^*$ (conjugate symmetry)
- (iv) $(\mathbf{u}, \mathbf{u}) \geq 0$, with equality if and only if $\mathbf{u} = \mathbf{0}$ (positive definiteness)

A vector space equipped with an inner product is called an *inner product space*.

Definition (Norm). The *norm* induced by an inner product is defined by $\|\mathbf{u}\| = \sqrt{(\mathbf{u}, \mathbf{u})}$. This provides the notions of length and distance in the vector space.

Note that positive definiteness is well-defined: conjugate symmetry implies $(\mathbf{u}, \mathbf{u}) = (\mathbf{u}, \mathbf{u})^*$, so $(\mathbf{u}, \mathbf{u}) \in \mathbb{R}$.

Proposition (Anti-linearity in first argument). For any inner product, we have

$$(\lambda \mathbf{u}, \mathbf{v}) = \lambda^*(\mathbf{u}, \mathbf{v})$$

for all $\mathbf{u}, \mathbf{v} \in V$ and $\lambda \in \mathbb{C}$.

Proof. Using conjugate symmetry and linearity in the second argument:

$$(\lambda \mathbf{u}, \mathbf{v}) = (\mathbf{v}, \lambda \mathbf{u})^* = (\lambda(\mathbf{v}, \mathbf{u}))^* = \lambda^*(\mathbf{v}, \mathbf{u})^* = \lambda^*(\mathbf{u}, \mathbf{v}). \quad \square$$

Definition (Basis). A set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ forms a *basis* of V if every $\mathbf{u} \in V$ can be uniquely written as a linear combination

$$\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$$

for some scalars $\lambda_i \in \mathbb{C}$. The *dimension* of a vector space is the number of elements in any basis.

Definition (Orthogonal and orthonormal bases). Let V be an inner product space with basis $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$.

- The basis is *orthogonal* if $(\mathbf{v}_i, \mathbf{v}_j) = 0$ for all $i \neq j$.
- The basis is *orthonormal* if it is orthogonal and $(\mathbf{v}_i, \mathbf{v}_i) = 1$ for all i .

Equivalently, an orthonormal basis satisfies $(\mathbf{v}_i, \mathbf{v}_j) = \delta_{ij}$, where δ_{ij} is the Kronecker delta.

Orthonormal bases are particularly convenient because they allow explicit computation of expansion coefficients.

Proposition (Expansion coefficients in orthonormal basis). Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an orthonormal basis of an inner product space V . For any $\mathbf{u} \in V$, the expansion

$$\mathbf{u} = \sum_{i=1}^n \lambda_i \mathbf{v}_i$$

has coefficients given by

$$\lambda_i = (\mathbf{v}_i, \mathbf{u}).$$

Proof. Taking the inner product of both sides with $\mathbf{v}_j$:

$$(\mathbf{v}_j, \mathbf{u}) = \left(\mathbf{v}_j, \sum_{i=1}^n \lambda_i \mathbf{v}_i \right) = \sum_{i=1}^n \lambda_i (\mathbf{v}_j, \mathbf{v}_i) = \sum_{i=1}^n \lambda_i \delta_{ji} = \lambda_j,$$

where we used linearity and additivity in the second argument, together with orthonormality. $\square$

1.1 Function spaces

The preceding definitions apply to finite-dimensional spaces such as $\mathbb{R}^n$ or $\mathbb{C}^n$. However, *functions* can also form vector spaces, typically of infinite dimension.

Definition (Function space). Let $\Omega \subseteq \mathbb{R}^n$ be a domain. The set of complex-valued functions $f : \Omega \rightarrow \mathbb{C}$ forms a vector space under pointwise operations:

- Addition: $(f + g)(x) = f(x) + g(x)$
- Scalar multiplication: $(\lambda f)(x) = \lambda f(x)$

for all $x \in \Omega$, $\lambda \in \mathbb{C}$.

Proposition (Solution spaces of linear equations). The set of solutions to a linear differential equation forms a vector space.

Proof. Let $\mathcal{L}$ be a linear differential operator. If f and g satisfy $\mathcal{L}f = 0$ and $\mathcal{L}g = 0$, then for any scalars α, β :

$$\mathcal{L}(\alpha f + \beta g) = \alpha \mathcal{L}f + \beta \mathcal{L}g = 0.$$

Hence $\alpha f + \beta g$ is also a solution, and the solution set is closed under linear combinations. $\square$

To apply the techniques of inner product spaces to functions, we need an appropriate inner product.

Definition (Inner product on function spaces). Let $\Omega \subseteq \mathbb{R}^n$ be a domain with measure $d\mu$. For functions $f, g : \Omega \rightarrow \mathbb{C}$, define

$$(f, g) = \int_{\Omega} f(x)^* g(x) d\mu.$$

This defines an inner product on the space of functions for which the integral converges.

This definition is the natural analogue of the finite-dimensional inner product $(\mathbf{u}, \mathbf{v}) = \sum_i u_i^* v_i$: viewing a function as a “continuous list” of its values, the integral replaces the sum over components.

Example (Inner product on interval and disk).

(i) Let $\Omega = [a, b] \subseteq \mathbb{R}$. With the standard Lebesgue measure $d\mu = dx$:

$$(f, g) = \int_a^b f(x)^* g(x) dx.$$

(ii) Let $\Omega = D^2 = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ be the unit disk. In polar coordinates (r, θ) , the measure is $d\mu = r dr d\theta$, giving:

$$(f, g) = \int_0^1 \int_0^{2\pi} f(r, \theta)^* g(r, \theta) r dr d\theta.$$

Note that the measure $d\mu$ depends on the domain and coordinate system. In polar coordinates, the factor of r arises from the Jacobian of the coordinate transformation. More general weighted measures will appear in Section 3.

1.2 Boundary conditions

When Ω has a boundary $\partial\Omega$, we often restrict attention to functions satisfying prescribed conditions on $\partial\Omega$. For the solution space to remain a vector space, these boundary conditions must be compatible with linear combinations.

Definition (Homogeneous boundary conditions). A boundary condition is *homogeneous* if whenever f and g satisfy the boundary conditions, then so does $\lambda f + \mu g$ for any $\lambda, \mu \in \mathbb{C}$ (or $\mathbb{R}$).

Example (Homogeneous vs non-homogeneous boundary conditions). Let $\Sigma = [a, b]$. We could require that $f(a) + 7f'(b) = 0$, or maybe $f(a) + 3f''(a) = 0$. These are examples of homogeneous boundary conditions. On the other hand, the requirement $f(a) = 1$ is *not* homogeneous.

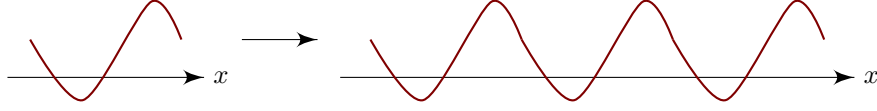
2 Fourier series

In this section, we develop the theory of Fourier series for periodic functions.

Definition (Periodic function). A function $f : \mathbb{R} \rightarrow \mathbb{C}$ is *periodic* with period $R > 0$ if $f(x+R) = f(x)$ for all $x \in \mathbb{R}$. We call R the *fundamental period* if it is the smallest such positive number.

Remark. It is often convenient to view a periodic function with period 2π as a function $f : S^1 \rightarrow \mathbb{C}$ from the unit circle to $\mathbb{C}$, parametrized by an angle $\theta \in [0, 2\pi)$.

Periodic functions arise naturally in many applications. Moreover, they can be used to model functions on a bounded interval: given any $f : [0, 1] \rightarrow \mathbb{C}$, we can extend it periodically to all of $\mathbb{R}$ by defining $f(x+n) = f(x)$ for all $n \in \mathbb{Z}$.



2.1 Fourier series

As mentioned in the previous chapter, we want to find a set of “basis functions” for periodic functions. A natural choice is the exponential function $e^{in\theta}$, where $n \in \mathbb{Z}$. These functions have period 2π and are straightforward to integrate and differentiate.

Proposition (Orthogonality of complex exponentials). The set of functions $\{e^{in\theta} : n \in \mathbb{Z}\}$ forms an orthogonal set with respect to the inner product on $[-\pi, \pi]$. Specifically, for any $m, n \in \mathbb{Z}$,

$$(e^{im\theta}, e^{in\theta}) = \int_{-\pi}^{\pi} e^{-im\theta} e^{in\theta} d\theta = \int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta = \begin{cases} 2\pi & n = m \\ 0 & n \neq m \end{cases} = 2\pi\delta_{nm}.$$

Consequently, the set $\left\{\frac{1}{\sqrt{2\pi}}e^{in\theta} : n \in \mathbb{Z}\right\}$ is orthonormal.

Proof. For $n = m$, the integrand is $e^{i \cdot 0 \cdot \theta} = 1$, so the integral evaluates to 2π . For $n \neq m$, we have

$$\int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta = \left[\frac{e^{i(n-m)\theta}}{i(n-m)} \right]_{-\pi}^{\pi} = \frac{e^{i(n-m)\pi} - e^{-i(n-m)\pi}}{i(n-m)} = 0,$$

since $e^{i(n-m)\pi} = e^{-i(n-m)\pi}$ when $n - m$ is an integer. $\square$

Fourier’s idea was to use this orthogonal set as a basis for *any* periodic function.

Definition (Fourier series and Fourier coefficients). Let $f : S^1 \rightarrow \mathbb{C}$ be a periodic function. The *Fourier series* of f is the formal series

$$f(\theta) = \sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta},$$

where the *Fourier coefficients* $\hat{f}_n$ are defined by

$$\hat{f}_n = \frac{1}{2\pi} (e^{in\theta}, f) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\theta} f(\theta) d\theta.$$

Remark. Strictly speaking, one should write $f(\theta) = \sum \hat{f}_n \frac{e^{in\theta}}{\sqrt{2\pi}}$ with $\hat{f}_n = \left(\frac{e^{in\theta}}{\sqrt{2\pi}}, f \right)$ to use the orthonormal basis. However, for convenience, we absorb all constant factors into the $\hat{f}_n$ coefficients.

For real-valued functions, there is an alternative formulation in terms of sines and cosines.

Proposition (Real Fourier series). Let $f : S^1 \rightarrow \mathbb{R}$ be a real-valued periodic function. Then the Fourier coefficients satisfy $\hat{f}_{-n} = \hat{f}_n^*$ (complex conjugate), and the Fourier series can be written as

$$f(\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\theta + b_n \sin n\theta),$$

where the real coefficients a_n and b_n are given by

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \cos n\theta \cdot f(\theta) d\theta, \quad b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \sin n\theta \cdot f(\theta) d\theta.$$

Proof. Since f is real-valued, we have

$$(\hat{f}_n)^* = \left(\frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\theta} f(\theta) d\theta \right)^* = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{in\theta} f(\theta) d\theta = \hat{f}_{-n}.$$

Thus we can rewrite the Fourier series as

$$f(\theta) = \hat{f}_0 + \sum_{n=1}^{\infty} (\hat{f}_n e^{in\theta} + \hat{f}_{-n} e^{-in\theta}) = \hat{f}_0 + \sum_{n=1}^{\infty} (\hat{f}_n e^{in\theta} + \hat{f}_n^* e^{-in\theta}).$$

Setting $\hat{f}_n = \frac{a_n - ib_n}{2}$ for $n \geq 1$ and using $e^{in\theta} + e^{-in\theta} = 2 \cos n\theta$ and $e^{in\theta} - e^{-in\theta} = 2i \sin n\theta$, we obtain the stated form. $\square$

Remark. When working with a real function, the choice between the complex exponential form and the sine/cosine form depends on the application. If f is odd (or even), the sine/cosine expansion is useful since the cosine (or sine) terms vanish. On the other hand, for differential

equations, the exponential form is often more convenient.

2.2 Convergence of Fourier series

When Fourier first proposed the idea of a Fourier series, mathematicians were skeptical. How can we be sure that the infinite series actually converges? It turns out that in many cases, it does not. To make this question precise, we first introduce some definitions.

Definition (Partial Fourier sum). Let f be a periodic function with Fourier coefficients $\hat{f}_m$. The n -th partial Fourier sum of f is

$$S_n f(\theta) = \sum_{m=-n}^n \hat{f}_m e^{im\theta}.$$

The central question is whether $S_n f$ “converges” to f as $n \rightarrow \infty$. There are several notions of convergence for functions, and we now define the most important ones.

Definition (Convergence in mean square). A sequence of functions (g_n) converges in mean square (or in norm) to a function g on $[-\pi, \pi]$ if

$$\lim_{n \rightarrow \infty} \|g_n - g\|^2 = \lim_{n \rightarrow \infty} \frac{1}{2\pi} \int_{-\pi}^{\pi} |g_n(\theta) - g(\theta)|^2 d\theta = 0.$$

Remark. Mean square convergence does not distinguish between functions that differ only on a set of measure zero. For instance, if $\lim_{n \rightarrow \infty} S_n f$ differs from f at countably many points (such as all rational points), mean square convergence still holds.

Definition (Pointwise convergence). A sequence of functions (g_n) converges pointwise to a function g if

$$\lim_{n \rightarrow \infty} g_n(\theta) = g(\theta)$$

for every θ in the domain.

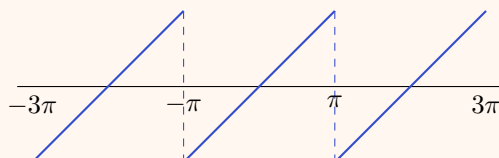
Definition (Uniform convergence). A sequence of functions (g_n) converges uniformly to a function g if

$$\lim_{n \rightarrow \infty} \sup_{\theta} |g_n(\theta) - g(\theta)| = 0.$$

This requires that the rate of convergence is independent of θ .

Different notions of convergence can yield different answers. A general criterion for convergence in norm requires more analysis (see the Part II Analysis of Functions course). Instead, we illustrate the behaviour with an example.

Example (Sawtooth function). Consider the *sawtooth function* $f(\theta) = \theta$ on $[-\pi, \pi]$, extended periodically to all of $\mathbb{R}$.



This function has jump discontinuities at odd multiples of π . We compute the Fourier coefficients. For $n = 0$:

$$\hat{f}_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} \theta \, d\theta = 0.$$

For $n \neq 0$, integrating by parts:

$$\hat{f}_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\theta} \theta \, d\theta = \left[-\frac{1}{2\pi in} e^{-in\theta} \theta \right]_{-\pi}^{\pi} + \frac{1}{2\pi in} \int_{-\pi}^{\pi} e^{-in\theta} \, d\theta = \frac{(-1)^{n+1}}{in}.$$

Thus the Fourier series is

$$f(\theta) = \sum_{n \neq 0} \frac{(-1)^{n+1}}{in} e^{in\theta}.$$

This series converges to $f(\theta)$ for all $\theta \neq (2m+1)\pi$, i.e., at every point where f is continuous.

Remark (Behaviour at discontinuities). At a discontinuity $\theta = \pi$, each term of the Fourier series vanishes (since $e^{-in\pi} = (-1)^n$), so the series converges to 0. This equals the average of the left and right limits:

$$\frac{1}{2} \left(\lim_{\varepsilon \rightarrow 0^+} f(\pi - \varepsilon) + \lim_{\varepsilon \rightarrow 0^+} f(-\pi + \varepsilon) \right) = \frac{1}{2}(\pi + (-\pi)) = 0.$$

This is a general phenomenon: at an isolated jump discontinuity, the Fourier series converges to the average of the one-sided limits.

2.3 Differentiability and Fourier series

The smoothness of a function is closely related to the decay of its Fourier coefficients. We begin with a motivating observation.

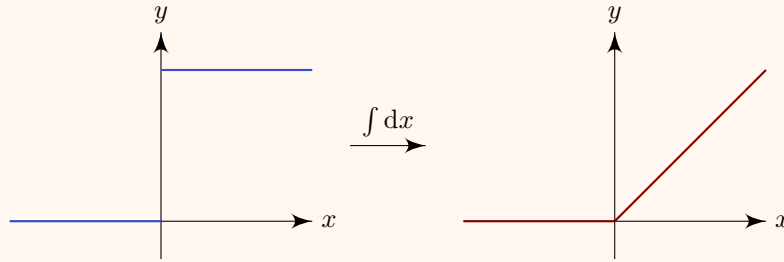
Example (Smoothing effect of integration). Consider the Heaviside step function

$$\Theta(x) = \begin{cases} 1 & x > 0 \\ 0 & x < 0 \end{cases}.$$

Its integral is

$$\int \Theta(x) \, dx = \begin{cases} x & x > 0 \\ 0 & x < 0 \end{cases},$$

which is now continuous. A further integration produces a function that is also differentiable.



Conversely, differentiation makes functions “rougher”: differentiating $\int \Theta(x) \, dx$ yields the discontinuous $\Theta(x)$, and differentiating again produces the Dirac δ -function.

This observation motivates characterizing the “smoothness” of a function by how many times it can be differentiated. The following theorem makes precise how smoothness affects Fourier coefficients.

Theorem (Decay of Fourier coefficients). Let $f : S^1 \rightarrow \mathbb{C}$ be a periodic function such that $f, f', \dots, f^{(m-1)}$ are all continuous, but $f^{(m)}$ has isolated discontinuities at points $\theta_1, \theta_2, \dots, \theta_r \in (-\pi, \pi)$. Then the Fourier coefficients satisfy

$$\hat{f}_k = O\left(\frac{1}{|k|^{m+1}}\right) \quad \text{as } |k| \rightarrow \infty.$$

In particular, smoother functions have faster-decaying Fourier coefficients.

Proof. For $k \neq 0$, we integrate by parts:

$$\begin{aligned} \hat{f}_k &= \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ik\theta} f(\theta) \, d\theta \\ &= \left[-\frac{1}{2\pi ik} e^{-ik\theta} f(\theta) \right]_{-\pi}^{\pi} + \frac{1}{2\pi ik} \int_{-\pi}^{\pi} e^{-ik\theta} f'(\theta) \, d\theta. \end{aligned}$$

The boundary term vanishes since f is continuous and periodic ($f(\pi) = f(-\pi)$). Repeating this process m times yields

$$\hat{f}_k = \frac{1}{(ik)^m} \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-ik\theta} f^{(m)}(\theta) \, d\theta.$$

Since $f^{(m)}$ has discontinuities, we split the integral around them. Writing $-\pi = \theta_0$ and $\pi = \theta_{r+1}$, and integrating by parts once more on each subinterval, we obtain boundary contributions from the jumps in $f^{(m)}$. The result is

$$\hat{f}_k = \frac{1}{(ik)^{m+1}} \frac{1}{2\pi} \left[\sum_{s=1}^r \left(f^{(m)}(\theta_s^+) - f^{(m)}(\theta_s^-) \right) e^{-ik\theta_s} + \int_{(-\pi, \pi) \setminus \{\theta_s\}} e^{-ik\theta} f^{(m+1)}(\theta) d\theta \right].$$

The sum is $O(1)$ and the integral is bounded, so $\hat{f}_k = O(k^{-(m+1)})$. $\square$

Remark. The intuition is clear: a smooth function varies slowly, so it is well-approximated by low-frequency Fourier modes, and higher-frequency coefficients decay rapidly. A rough function with rapid oscillations requires higher-frequency terms, so the coefficients decay more slowly.

An important result relating the inner product of a function to its Fourier coefficients is Parseval's theorem.

Theorem (Parseval's theorem). Let $f : S^1 \rightarrow \mathbb{C}$ be a periodic function with Fourier coefficients $\hat{f}_n$. Then

$$(f, f) = \int_{-\pi}^{\pi} |f(\theta)|^2 d\theta = 2\pi \sum_{n \in \mathbb{Z}} |\hat{f}_n|^2.$$

Proof. Substituting the Fourier expansion and using orthogonality:

$$\begin{aligned} (f, f) &= \int_{-\pi}^{\pi} |f(\theta)|^2 d\theta \\ &= \int_{-\pi}^{\pi} \left(\sum_{m \in \mathbb{Z}} \hat{f}_m^* e^{-im\theta} \right) \left(\sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta} \right) d\theta \\ &= \sum_{m, n \in \mathbb{Z}} \hat{f}_m^* \hat{f}_n \int_{-\pi}^{\pi} e^{i(n-m)\theta} d\theta \\ &= 2\pi \sum_{m, n \in \mathbb{Z}} \hat{f}_m^* \hat{f}_n \delta_{mn} \\ &= 2\pi \sum_{n \in \mathbb{Z}} |\hat{f}_n|^2. \end{aligned} \quad \square$$

Remark. This proof assumes that the Fourier series converges to f and that the sums can be interchanged with the integral. A rigorous treatment requires analysis beyond the scope of this course.

Parseval's theorem has surprising applications in number theory.

Definition (Riemann zeta function). The *Riemann zeta function* is

defined for $s > 1$ by

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

Example (Computing $\zeta(2)$ via Parseval). We show that $\zeta(2) = \frac{\pi^2}{6}$. From the sawtooth example, the function $f(\theta) = \theta$ on $[-\pi, \pi]$ has Fourier coefficients

$$\hat{f}_0 = 0, \quad \hat{f}_n = \frac{i(-1)^n}{n} \text{ for } n \neq 0.$$

By direct computation,

$$(f, f) = \int_{-\pi}^{\pi} \theta^2 \, d\theta = \frac{2\pi^3}{3}.$$

On the other hand, Parseval's theorem gives

$$(f, f) = 2\pi \sum_{n \in \mathbb{Z}} |\hat{f}_n|^2 = 2\pi \cdot 2 \sum_{n=1}^{\infty} \frac{1}{n^2} = 4\pi \sum_{n=1}^{\infty} \frac{1}{n^2}.$$

Equating these expressions yields

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \zeta(2) = \frac{\pi^2}{6}.$$

By integrating the sawtooth function repeatedly and applying the same argument, one can show that $\zeta(2m) = \pi^{2m} q$ for some $q \in \mathbb{Q}$ for all positive integers m .

Remark. One may ask: why use $\{e^{in\theta}\}$ as a basis? In finite dimensions, any linearly independent set forms a basis, but in practice we choose bases that respect the symmetries of the problem. For periodic functions, the complex exponentials $e^{in\theta}$ are natural because they are eigenfunctions of the differentiation operator and capture the periodicity of the domain.

3 Sturm-Liouville Theory

3.1 Sturm-Liouville operators

In finite dimensions, a linear map $M : V \rightarrow W$ between inner product spaces with bases $\{\mathbf{v}_i\}$ and $\{\mathbf{w}_a\}$ can be represented by a matrix with entries $M_{ai} = (\mathbf{w}_a, M\mathbf{v}_i)$. A map $M : V \rightarrow V$ is called *self-adjoint* if $M^\dagger = M$ as matrices. To extend this notion to infinite-dimensional spaces where matrix representations may not exist, we use the following definition.

Definition (Adjoint and self-adjoint operator). Let V be an inner product space and $A : V \rightarrow V$ a linear map. The *adjoint* of A is a map $A^\dagger : V \rightarrow V$ such that

$$(A^\dagger \mathbf{u}, \mathbf{v}) = (\mathbf{u}, A\mathbf{v})$$

for all $\mathbf{u}, \mathbf{v} \in V$. The map A is *self-adjoint* (or *Hermitian*) if $A = A^\dagger$, i.e.,

$$(A\mathbf{u}, \mathbf{v}) = (\mathbf{u}, A\mathbf{v})$$

for all $\mathbf{u}, \mathbf{v} \in V$.

Self-adjoint operators have special spectral properties. Recall that an *eigenvalue* λ of M is a root of $\det(M - \lambda I) = 0$, and a corresponding *eigenvector* $\mathbf{v}$ satisfies $M\mathbf{v} = \lambda\mathbf{v}$.

Proposition (Self-adjoint operators have real eigenvalues). Let $M : V \rightarrow V$ be a self-adjoint operator. Then all eigenvalues of M are real.

Proof. Suppose $M\mathbf{v} = \lambda\mathbf{v}$ for some nonzero eigenvector $\mathbf{v}$. Then

$$\lambda(\mathbf{v}, \mathbf{v}) = (\mathbf{v}, M\mathbf{v}) = (M\mathbf{v}, \mathbf{v}) = \lambda^*(\mathbf{v}, \mathbf{v}).$$

Since $(\mathbf{v}, \mathbf{v}) \neq 0$, we have $\lambda = \lambda^*$, so $\lambda \in \mathbb{R}$. □

Proposition (Eigenvectors of self-adjoint operators are orthogonal). Let $M : V \rightarrow V$ be a self-adjoint operator. Eigenvectors corresponding to distinct eigenvalues are orthogonal.

Proof. Suppose $M\mathbf{v}_i = \lambda_i\mathbf{v}_i$ and $M\mathbf{v}_j = \lambda_j\mathbf{v}_j$ with $\lambda_i \neq \lambda_j$. Then

$$\lambda_i(\mathbf{v}_j, \mathbf{v}_i) = (\mathbf{v}_j, M\mathbf{v}_i) = (M\mathbf{v}_j, \mathbf{v}_i) = \lambda_j(\mathbf{v}_j, \mathbf{v}_i).$$

Since $\lambda_i \neq \lambda_j$, we must have $(\mathbf{v}_j, \mathbf{v}_i) = 0$. □

Eigenvector expansions provide a systematic method for solving linear equations.

Example (Solving linear equations via eigenvector expansion). Consider the equation $M\mathbf{u} = \mathbf{f}$, where M is self-adjoint with eigenvectors $\{\mathbf{v}_i\}$

forming a basis. We can expand $\mathbf{u} = \sum_i u_i \mathbf{v}_i$ and $\mathbf{f} = \sum_i f_i \mathbf{v}_i$. Then

$$M\mathbf{u} = \sum_i u_i M\mathbf{v}_i = \sum_i u_i \lambda_i \mathbf{v}_i = \sum_i f_i \mathbf{v}_i.$$

By orthogonality, $u_i \lambda_i = f_i$, so

$$u_i = \frac{f_i}{\lambda_i},$$

provided all $\lambda_i \neq 0$.

Sturm–Liouville theory extends these ideas to infinite-dimensional function spaces, where “matrices” become differential operators.

Definition (Linear differential operator). A *linear differential operator* of order p on an interval $[a, b]$ is an operator of the form

$$\mathcal{L} = A_p(x) \frac{d^p}{dx^p} + A_{p-1}(x) \frac{d^{p-1}}{dx^{p-1}} + \cdots + A_1(x) \frac{d}{dx} + A_0(x),$$

where $A_0, A_1, \dots, A_p : [a, b] \rightarrow \mathbb{C}$ are coefficient functions and $A_p(x) \neq 0$. Linearity means $\mathcal{L}(\alpha f + \beta g) = \alpha \mathcal{L}f + \beta \mathcal{L}g$ for all scalars α, β and functions f, g .

In most applications, we consider second-order operators ($p = 2$). The key question is: when is such an operator self-adjoint?

Proposition (Reduction to Sturm–Liouville form). Any second-order linear differential operator

$$\mathcal{L}y = P(x) \frac{d^2 y}{dx^2} + R(x) \frac{dy}{dx} + Q(x)y$$

with $P(x) \neq 0$ can be written (up to multiplication by a nonzero function) in *Sturm–Liouville form*.

Proof. Factor out P and complete the derivative:

$$\begin{aligned} \mathcal{L}y &= P \frac{d^2 y}{dx^2} + R \frac{dy}{dx} + Qy \\ &= P \left[\frac{d^2 y}{dx^2} + \frac{R}{P} \frac{dy}{dx} + \frac{Q}{P} y \right] \\ &= P \left[e^{-\int \frac{R}{P} dx} \frac{d}{dx} \left(e^{\int \frac{R}{P} dx} \frac{dy}{dx} \right) + \frac{Q}{P} y \right]. \end{aligned}$$

Setting $p(x) = \exp \left(\int \frac{R}{P} dx \right)$ and $q(x) = -\frac{Q}{P}p(x)$, and absorbing the factor Pp^{-1} , we obtain

$$\mathcal{L} = \frac{d}{dx} \left(p(x) \frac{d}{dx} \right) - q(x). \quad \square$$

Definition (Sturm–Liouville operator). A *Sturm–Liouville operator* on $[a, b]$ is a second-order differential operator of the form

$$\mathcal{L} = \frac{d}{dx} \left(p(x) \frac{d}{dx} \right) - q(x),$$

where $p, q : [a, b] \rightarrow \mathbb{R}$ are real-valued functions with $p(x) > 0$ on (a, b) .

Proposition (Self-adjointness of Sturm–Liouville operators). A Sturm–Liouville operator $\mathcal{L}$ satisfies

$$(f, \mathcal{L}g) = [(f^*g' - f'^*g)p]_a^b + (\mathcal{L}f, g)$$

for all sufficiently differentiable functions f, g . In particular, $\mathcal{L}$ is self-adjoint if the boundary term vanishes.

Proof. Integrating by parts twice:

$$\begin{aligned} (f, \mathcal{L}g) &= \int_a^b f^* \left(\frac{d}{dx} \left(p \frac{dg}{dx} \right) - qg \right) dx \\ &= [f^*pg']_a^b - \int_a^b \left(\frac{df^*}{dx} p \frac{dg}{dx} + f^*qg \right) dx \\ &= [f^*pg' - f'^*pg]_a^b + \int_a^b \left(\frac{d}{dx} \left(p \frac{df^*}{dx} \right) - qf^* \right) g dx \\ &= [(f^*g' - f'^*g)p]_a^b + (\mathcal{L}f, g). \end{aligned} \quad \square$$

Remark. The boundary term $[(f^*g' - f'^*g)p]_a^b$ vanishes in several important cases:

- (i) periodic boundary conditions: $f(a) = f(b)$ and $f'(a) = f'(b)$;
- (ii) Dirichlet conditions: $f(a) = f(b) = 0$;
- (iii) singular endpoints: $p(a) = 0$ or $p(b) = 0$.

Example (Second derivative operator). The operator $\mathcal{L} = \frac{d^2}{dx^2}$ corresponds to $p = 1$ and $q = 0$. For periodic functions on $[a, b]$, the boundary terms cancel, giving

$$\int_a^b f^* \frac{d^2g}{dx^2} dx = \int_a^b \frac{d^2f^*}{dx^2} g dx.$$

Self-adjointness requires a *second-order* operator; a first-order operator would acquire a sign change from a single integration by parts.

Just as in finite dimensions, self-adjoint operators have eigenfunctions and eigenvalues with special properties. To state these results, we introduce a

weighted inner product.

Definition (Weighted inner product). Let $w : [a, b] \rightarrow \mathbb{R}$ be a *weight function*: a real, non-negative function with at most finitely many zeros. The *weighted inner product* is

$$(f, g)_w = \int_a^b f^*(x)g(x)w(x) \, dx.$$

Remark. Weight functions arise naturally in curvilinear coordinates. For instance, in polar coordinates on a disk, the area element is $r \, dr \, d\theta$, giving weight $w(r) = r$ that vanishes at the origin. We allow finitely many zeros so that $(f, f)_w = 0$ still implies $f = 0$ for continuous f .

Definition (Eigenfunction of a Sturm–Liouville operator). Let $\mathcal{L}$ be a Sturm–Liouville operator on $[a, b]$ and w a weight function. An *eigenfunction* of $\mathcal{L}$ with weight w is a nonzero function $y : [a, b] \rightarrow \mathbb{C}$ satisfying

$$\mathcal{L}y = \lambda wy$$

for some $\lambda \in \mathbb{C}$, called the *eigenvalue*. The eigenfunction must also satisfy appropriate boundary conditions.

Remark. The eigenvalue equation $\mathcal{L}y = \lambda wy$ is not satisfied by arbitrary functions. The combination of the differential equation, boundary conditions, and constraints on w (positive, real, finitely many zeros) severely restricts which y are eigenfunctions. The weight function w is particularly convenient for problems in curvilinear coordinates.

Proposition (Sturm-Liouville eigenvalues, reality). The eigenvalues of a Sturm-Liouville operator are real.

Proof. Suppose $\mathcal{L}y_i = \lambda_i wy_i$. Then

$$\lambda_i(y_i, y_i)_w = \lambda_i(y_i, wy_i) = (y_i, \mathcal{L}y_i) = (\mathcal{L}y_i, y_i) = (\lambda_i wy_i, y_i) = \lambda_i^*(y_i, y_i)_w.$$

Since $(y_i, y_i)_w \neq 0$, we have $\lambda_i = \lambda_i^*$. $\square$

Note that the first and last terms use the weighted inner product, but the middle terms use the unweighted inner product.

Proposition (Sturm-Liouville eigenfunctions, orthogonality). Eigenfunctions with different eigenvalues (but same weight) are orthogonal.

Proof. Let $\mathcal{L}y_i = \lambda_i wy_i$ and $\mathcal{L}y_j = \lambda_j wy_j$. Then

$$\lambda_i(y_j, y_i)_w = (y_j, \mathcal{L}y_i) = (\mathcal{L}y_j, y_i) = \lambda_j(y_j, y_i)_w.$$

Since $\lambda_i \neq \lambda_j$, we must have $(y_j, y_i)_w = 0$. $\square$

Those were pretty straightforward manipulations. However, the main results of Sturm–Liouville theory are significantly harder, and we will not prove them. We shall just state them and explore some examples.

Theorem (Sturm-Liouville spectrum, compact domain). On a *compact* domain, the eigenvalues $\lambda_1, \lambda_2, \dots$ form a countably infinite sequence and are discrete.

This will be a rather helpful result in quantum mechanics, since in quantum mechanics, the possible values of, say, the energy are the eigenvalues of the Hamiltonian operator. Then this result says that the possible values of the energy are discrete and form an infinite sequence.

Note here the word *compact*. In quantum mechanics, if we restrict a particle in a well $[0, 1]$, then it will have quantized energy level since the domain is compact. However, if the particle is free, then it can have any energy at all since we no longer have a compact domain. Similarly, angular momentum is quantized, since it describes rotations, which takes values in S^1 , which is compact.

Theorem (Sturm-Liouville completeness). The eigenfunctions are complete: any function $f : [a, b] \rightarrow \mathbb{C}$ (obeying appropriate boundary conditions) can be expanded as

$$f(x) = \sum_n \hat{f}_n y_n(x),$$

where

$$\hat{f}_n = \int y_n^*(x) f(x) w(x) \, dx.$$

Example (Fourier series as eigenfunction expansion). Consider $[a, b] = [-L, L]$ with $\mathcal{L} = \frac{d^2}{dx^2}$, $w = 1$, and periodic boundary conditions. The eigenvalue equation

$$\frac{d^2 y_n}{dx^2} = \lambda_n y_n(x)$$

has eigenfunctions

$$y_n(x) = \exp\left(\frac{in\pi x}{L}\right), \quad n \in \mathbb{Z},$$

with eigenvalues

$$\lambda_n = -\frac{n^2 \pi^2}{L^2}.$$

This recovers the Fourier series as a special case of Sturm–Liouville theory.

Example (Hermite polynomials). Consider the differential equation on $\mathbb{R}$:

$$\frac{1}{2}H'' - xH' = \lambda H.$$

To put this in Sturm-Liouville form, we compute $p(x) = \exp(-\int_0^x 2t \, dt) = e^{-x^2}$ and $q(x) = 0$, giving

$$\frac{d}{dx} \left(e^{-x^2} \frac{dH}{dx} \right) = -2\lambda e^{-x^2} H(x).$$

The weight function is $w(x) = e^{-x^2}$.

For self-adjointness on the unbounded domain, we require $e^{-x^2} H(x)^2 \rightarrow 0$ as $|x| \rightarrow \infty$, which ensures the boundary terms vanish.

The eigenfunctions are the *Hermite polynomials*:

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} (e^{-x^2}), \quad n = 0, 1, 2, \dots$$

These are indeed polynomials: each differentiation of e^{-x^2} produces terms containing e^{-x^2} , which cancels with the prefactor e^{x^2} .

Just as for matrices, eigenfunction expansions provide a systematic method for solving forced differential equations.

Proposition (Solving forced equations via eigenfunction expansion). Consider the equation $\mathcal{L}g = w(x)F(x)$, where $\mathcal{L}$ is a Sturm-Liouville operator with eigenfunctions $\{y_n\}$ and eigenvalues $\{\lambda_n\}$. If all $\lambda_n \neq 0$, then

$$g(x) = \sum_n \frac{\hat{F}_n}{\lambda_n} y_n(x),$$

where $\hat{F}_n = (y_n, F)_w$ are the expansion coefficients of F .

Proof. Expand $g(x) = \sum_n \hat{g}_n y_n(x)$ and $F(x) = \sum_n \hat{F}_n y_n(x)$. By linearity of $\mathcal{L}$:

$$\mathcal{L}g = \sum_n \hat{g}_n \mathcal{L}y_n = \sum_n \hat{g}_n \lambda_n w(x) y_n(x).$$

Comparing with $w(x)F(x) = w(x) \sum_n \hat{F}_n y_n(x)$ and using orthogonality, we get $\hat{g}_n \lambda_n = \hat{F}_n$, so $\hat{g}_n = \hat{F}_n / \lambda_n$. $\square$

This can be reformulated using a Green's function.

Definition (Green's function). Let $\mathcal{L}$ be a Sturm-Liouville operator with eigenfunctions $\{y_n\}$, eigenvalues $\{\lambda_n\}$, and weight w . The *Green's function* for $\mathcal{L}$ is

$$G(x, t) = \sum_n \frac{1}{\lambda_n} y_n^*(t) y_n(x),$$

provided all $\lambda_n \neq 0$.

Proposition (Solution via Green's function). The solution to $\mathcal{L}g = w(x)F(x)$ can be written as

$$g(x) = \int_a^b G(x, t)F(t)w(t) \, dt.$$

Proof. Using $\hat{F}_n = (y_n, F)_w = \int_a^b y_n^*(t)F(t)w(t) \, dt$:

$$g(x) = \sum_n \frac{\hat{F}_n}{\lambda_n} y_n(x) = \int_a^b \left(\sum_n \frac{1}{\lambda_n} y_n^*(t) y_n(x) \right) F(t)w(t) \, dt = \int_a^b G(x, t)F(t)w(t) \, dt,$$

where we have interchanged sum and integral (justified under appropriate convergence conditions). $\square$

Remark. The Green's function $G(x, t)$ depends only on the operator $\mathcal{L}$ (through its eigenfunctions and eigenvalues), not on the forcing term F . It acts as the “inverse” of $\mathcal{L}$, analogous to the inverse matrix. Just as a matrix inverse exists when all eigenvalues are nonzero, the Green's function requires $\lambda_n \neq 0$ for all n .

The weighted generalization of Parseval's theorem holds for Sturm–Liouville eigenfunctions.

Theorem (Parseval's theorem for weighted inner products). Let $\{y_n\}$ be orthonormal eigenfunctions of a Sturm–Liouville operator with weight w . For any function f with expansion $f = \sum_n \hat{f}_n y_n$, we have

$$(f, f)_w = \sum_n |\hat{f}_n|^2.$$

Proof. Substituting the eigenfunction expansion:

$$\begin{aligned} (f, f)_w &= \int_a^b f^*(x)f(x)w(x) \, dx \\ &= \sum_{n,m} \hat{f}_n^* \hat{f}_m \int_a^b y_n^*(x)y_m(x)w(x) \, dx \\ &= \sum_{n,m} \hat{f}_n^* \hat{f}_m (y_n, y_m)_w \\ &= \sum_n |\hat{f}_n|^2, \end{aligned}$$

using orthonormality $(y_n, y_m)_w = \delta_{nm}$. $\square$

3.2 Least squares approximation

In practice, we must truncate infinite eigenfunction expansions. The following theorem shows that the standard expansion coefficients give the optimal truncation.

Theorem (Least squares approximation). Let $f : [a, b] \rightarrow \mathbb{C}$ and let $\{y_1, y_2, \dots, y_N\}$ be orthonormal eigenfunctions with respect to weight w . Among all approximations of the form

$$g(x) = \sum_{k=1}^N c_k y_k(x),$$

the weighted mean-square error $\|f - g\|_w^2 = (f - g, f - g)_w$ is minimized when

$$c_k = \hat{f}_k = (y_k, f)_w,$$

i.e., when the coefficients are the usual expansion coefficients.

Proof. Expanding the error:

$$(f - g, f - g)_w = (f, f)_w + (g, g)_w - (f, g)_w - (g, f)_w.$$

Using orthonormality and $g = \sum_k c_k y_k$, we have $(g, g)_w = \sum_k |c_k|^2$, $(f, g)_w = \sum_k \hat{f}_k^* c_k$, and $(g, f)_w = \sum_k c_k^* \hat{f}_k$.

To minimize, we differentiate with respect to c_j (treating c_j and c_j^* as independent, which can be justified by varying real and imaginary parts separately):

$$\frac{\partial}{\partial c_j} (f - g, f - g)_w = c_j^* - \hat{f}_j^* = 0.$$

Thus $c_j = \hat{f}_j$ at the critical point. The second derivative $\frac{\partial^2}{\partial c_i^* \partial c_j} (f - g, f - g)_w = \delta_{ij} > 0$ confirms this is a minimum. $\square$

Remark. This result shows that truncating the eigenfunction expansion at any point gives the best possible approximation using that many terms. There is no better choice of coefficients.

4 Partial differential equations

In this section, we apply the theory developed so far to solve partial differential equations (PDEs). The primary technique is *separation of variables*, which reduces a PDE to a system of ordinary differential equations.

4.1 Laplace's equation

Definition (Laplacian and Laplace's equation). The *Laplacian* (or *Laplace operator*) on $\mathbb{R}^n$ is the differential operator

$$\nabla^2 = \sum_{i=1}^n \frac{\partial^2}{\partial x_i^2}.$$

Laplace's equation for a function $\phi : \Omega \subseteq \mathbb{R}^n \rightarrow \mathbb{C}$ is

$$\nabla^2 \phi = 0.$$

Definition (Harmonic function). A function satisfying Laplace's equation is called *harmonic*.

Remark. Laplace's equation arises in many physical contexts:

- For a conservative force $\mathbf{F} = -\nabla\phi$ in a source-free region where $\nabla \cdot \mathbf{F} = 0$ (e.g., electrostatics without charges).
- As the steady-state limit of the heat equation $\frac{\partial\phi}{\partial t} = \kappa\nabla^2\phi$ when $\frac{\partial\phi}{\partial t} = 0$.
- As the time-independent case of the wave equation $\frac{\partial^2\phi}{\partial t^2} = c^2\nabla^2\phi$.

Definition (Dirichlet and Neumann boundary conditions). Let Ω be a domain with boundary $\partial\Omega$. A *Dirichlet boundary condition* specifies the value of ϕ on $\partial\Omega$:

$$\phi|_{\partial\Omega} = f.$$

A *Neumann boundary condition* specifies the normal derivative:

$$\mathbf{n} \cdot \nabla\phi|_{\partial\Omega} = g.$$

Theorem (Uniqueness for the Dirichlet problem). Let $\Omega \subseteq \mathbb{R}^n$ be a compact domain with boundary $\partial\Omega$, and let $f : \partial\Omega \rightarrow \mathbb{R}$. Then there is at most one solution to Laplace's equation $\nabla^2\phi = 0$ on Ω with $\phi|_{\partial\Omega} = f$.

Proof. Suppose ϕ_1 and ϕ_2 are both solutions with $\phi_1|_{\partial\Omega} = \phi_2|_{\partial\Omega} = f$. Let

$\Phi = \phi_1 - \phi_2$, so $\Phi = 0$ on $\partial\Omega$ and $\nabla^2\Phi = 0$ in Ω . Then

$$0 = \int_{\Omega} \Phi \nabla^2 \Phi \, d^n x = - \int_{\Omega} |\nabla \Phi|^2 \, d^n x + \int_{\partial\Omega} \Phi (\nabla \Phi \cdot \mathbf{n}) \, d^{n-1} x.$$

The boundary term vanishes since $\Phi = 0$ on $\partial\Omega$, leaving

$$\int_{\Omega} |\nabla \Phi|^2 \, d^n x = 0.$$

Since $|\nabla \Phi|^2 \geq 0$ with equality if and only if $\nabla \Phi = 0$, we conclude Φ is constant. Since $\Phi = 0$ on $\partial\Omega$, we have $\Phi = 0$ throughout, i.e., $\phi_1 = \phi_2$. $\square$

Remark. For Neumann boundary conditions, uniqueness holds up to an additive constant.

4.2 Laplace's equation in the unit disk in $\mathbb{R}^2$

Example (Laplace's equation on the unit disk). Let $\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$ be the closed unit disk. In two dimensions, Laplace's equation can be written using complex coordinates $z = x + iy$ and $\bar{z} = x - iy$ as

$$\nabla^2 \phi = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 4 \frac{\partial^2 \phi}{\partial z \partial \bar{z}} = 0.$$

The general solution is therefore $\phi(z, \bar{z}) = \psi(z) + \chi(\bar{z})$ for holomorphic functions ψ and χ .

Suppose we seek a solution satisfying the Dirichlet condition $\phi|_{\partial\Omega} = f(\theta)$, where θ parametrizes the boundary circle. We expand f as a Fourier series:

$$f(\theta) = \sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta} = \hat{f}_0 + \sum_{n=1}^{\infty} \hat{f}_n e^{in\theta} + \sum_{n=1}^{\infty} \hat{f}_{-n} e^{-in\theta}.$$

On the boundary, $z = e^{i\theta}$ and $\bar{z} = e^{-i\theta}$, so we can write

$$f(\theta) = \hat{f}_0 + \sum_{n=1}^{\infty} \hat{f}_n z^n + \sum_{n=1}^{\infty} \hat{f}_{-n} \bar{z}^n.$$

This suggests defining the solution on the entire disk by

$$\phi(z, \bar{z}) = \hat{f}_0 + \sum_{n=1}^{\infty} \hat{f}_n z^n + \sum_{n=1}^{\infty} \hat{f}_{-n} \bar{z}^n.$$

This function:

- (i) satisfies the boundary condition by construction;
- (ii) converges for $|z| < 1$ whenever the Fourier series for f converges;
- (iii) has the form $\psi(z) + \chi(\bar{z})$, hence solves Laplace's equation.

By uniqueness, this is the solution.

4.3 Separation of variables

The complex variable method is specific to $\mathbb{R}^2$. In higher dimensions, we use *separation of variables*: we seek solutions of the form $\phi = X_1(x_1)X_2(x_2) \cdots X_n(x_n)$, where each factor depends on only one variable.

Example (Laplace's equation on a half-infinite rectangular prism). Let $\Omega = \{(x, y, z) \in \mathbb{R}^3 : 0 \leq x \leq a, 0 \leq y \leq b, z \geq 0\}$. We seek a solution ψ to $\nabla^2 \psi = 0$ with boundary conditions:

$$\begin{aligned} \psi(0, y, z) = \psi(a, y, z) &= 0, & (\text{homogeneous in } x) \\ \psi(x, 0, z) = \psi(x, b, z) &= 0, & (\text{homogeneous in } y) \\ \lim_{z \rightarrow \infty} \psi(x, y, z) &= 0, & (\text{decay at infinity}) \\ \psi(x, y, 0) &= f(x, y), & (\text{inhomogeneous at } z = 0) \end{aligned}$$

where f is a given function.

Step 1: Separation ansatz. We try $\psi(x, y, z) = X(x)Y(y)Z(z)$. Substituting into $\nabla^2 \psi = 0$:

$$YZX'' + XZY'' + XYZ'' = 0 \implies \frac{X''}{X} + \frac{Y''}{Y} + \frac{Z''}{Z} = 0.$$

Since each term depends on only one variable, and the sum is constant (zero), each term must be separately constant:

$$X'' = -\lambda X, \quad Y'' = -\mu Y, \quad Z'' = (\lambda + \mu)Z.$$

The signs are chosen for convenience.

Step 2: Solve the ODEs. The general solutions are

$$\begin{aligned} X &= a \sin \sqrt{\lambda}x + b \cos \sqrt{\lambda}x, \\ Y &= c \sin \sqrt{\mu}y + d \cos \sqrt{\mu}y, \\ Z &= g \exp(\sqrt{\lambda + \mu}z) + h \exp(-\sqrt{\lambda + \mu}z). \end{aligned}$$

Step 3: Apply homogeneous boundary conditions. We impose the conditions that ψ vanishes at the walls and at infinity:

- At $x = 0$: $\psi(0, y, z) = 0$ requires $b = 0$.
- At $x = a$: $\psi(a, y, z) = 0$ requires $\lambda = \left(\frac{n\pi}{a}\right)^2$ for $n \in \mathbb{N}$.
- At $y = 0$: $\psi(x, 0, z) = 0$ requires $d = 0$.
- At $y = b$: $\psi(x, b, z) = 0$ requires $\mu = \left(\frac{m\pi}{b}\right)^2$ for $m \in \mathbb{N}$.
- As $z \rightarrow \infty$: $\psi(x, y, z) \rightarrow 0$ requires $g = 0$.

Thus, for each $n, m \in \mathbb{N}$, we have a separated solution

$$\psi_{n,m}(x, y, z) = A_{n,m} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \exp(-s_{n,m}z),$$

where $A_{n,m}$ is an arbitrary constant and

$$s_{n,m}^2 = \left(\frac{n^2}{a^2} + \frac{m^2}{b^2} \right) \pi^2.$$

This satisfies the homogeneous boundary conditions but not yet the inhomogeneous condition at $z = 0$.

Step 4: Form general solution and determine coefficients. By linearity, the general solution satisfying all homogeneous boundary conditions is

$$\psi(x, y, z) = \sum_{n,m=1}^{\infty} A_{n,m} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \exp(-s_{n,m}z).$$

To satisfy the inhomogeneous condition $\psi(x, y, 0) = f(x, y)$, we need

$$\sum_{n,m=1}^{\infty} A_{n,m} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) = f(x, y). \quad (\dagger)$$

We use the orthogonality relation

$$\int_0^a \sin\left(\frac{k\pi x}{a}\right) \sin\left(\frac{n\pi x}{a}\right) dx = \delta_{k,n} \frac{a}{2}.$$

Multiplying $(\dagger)$ by $\sin\left(\frac{k\pi x}{a}\right)$ and integrating over $[0, a]$:

$$\sum_{n,m=1}^{\infty} A_{n,m} \int_0^a \sin\left(\frac{k\pi x}{a}\right) \sin\left(\frac{n\pi x}{a}\right) dx \sin\left(\frac{m\pi y}{b}\right) = \int_0^a \sin\left(\frac{k\pi x}{a}\right) f(x, y) dx.$$

Applying orthogonality:

$$\frac{a}{2} \sum_{m=1}^{\infty} A_{k,m} \sin\left(\frac{m\pi y}{b}\right) = \int_0^a \sin\left(\frac{k\pi x}{a}\right) f(x, y) dx.$$

Repeating with the y variable:

$$\frac{ab}{4} A_{k,j} = \int_{[0,a] \times [0,b]} \sin\left(\frac{k\pi x}{a}\right) \sin\left(\frac{j\pi y}{b}\right) f(x, y) dx dy. \quad (*)$$

Thus the solution is

$$\psi(x, y, z) = \sum_{n,m=1}^{\infty} A_{n,m} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \exp(-s_{n,m}z)$$

where $A_{m,n}$ is given by $(*)$. Uniqueness of the Dirichlet problem guarantees this is the solution.

Remark. If we had a boundary condition at finite z (say $0 \leq z \leq c$), both exponentials $\exp(\pm\sqrt{\lambda+\mu}z)$ would contribute. Similarly, if ψ does not vanish at the other boundaries, the cos terms would also contribute.

In general, computing the coefficients $A_{m,n}$ requires evaluating the integral in (*), which may be difficult.

Example (Rectangular prism with constant boundary). Suppose $f(x, y) = 1$. Then

$$A_{m,n} = \frac{4}{ab} \int_0^a \sin\left(\frac{n\pi x}{a}\right) dx \int_0^b \sin\left(\frac{m\pi y}{b}\right) dy = \begin{cases} \frac{16}{\pi^2 mn} & n, m \text{ both odd} \\ 0 & \text{otherwise} \end{cases}$$

Hence we have

$$\psi(x, y, z) = \frac{16}{\pi^2} \sum_{n,m \text{ odd}} \frac{1}{nm} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{b}\right) \exp(-s_{m,n}z).$$

4.4 Laplace's equation in spherical polar coordinates

For spherically symmetric domains, we use spherical coordinates.

Definition (Spherical coordinates). *Spherical coordinates* (r, θ, ϕ) on $\mathbb{R}^3$ are defined by

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta,$$

where $r \geq 0$, $0 \leq \theta \leq \pi$ (polar angle), and $0 \leq \phi < 2\pi$ (azimuthal angle). The Laplacian in these coordinates is

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2},$$

and the volume element is $dV = r^2 \sin \theta \, dr \, d\theta \, d\phi$.

Consider the domain $\Omega = \{(x, y, z) \in \mathbb{R}^3 : \sqrt{x^2 + y^2 + z^2} \leq a\}$. We restrict to *axisymmetric solutions* where $\psi(r, \theta, \phi) = \psi(r, \theta)$ does not depend on ϕ .

Applying separation of variables with $\psi(r, \theta) = R(r)\Theta(\theta)$, Laplace's equation becomes

$$\frac{\psi}{r^2} \left[\frac{1}{R} \frac{d}{dr} (r^2 R') + \frac{1}{\Theta \sin \theta} \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) \right] = 0.$$

Since each term depends on only one variable and the sum is constant (zero), each must be separately constant:

$$\frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = \lambda R, \quad \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) = -\lambda \sin \theta \Theta.$$

Both equations are eigenfunction equations for Sturm-Liouville operators. The angular equation has weight function $w(\theta) = \sin \theta$.

4.4.1 Legendre polynomials

Definition (Legendre's equation). The angular equation from separation of variables is *Legendre's equation*. Using the substitution $x = \cos \theta$ (where $x \in [-1, 1]$), it takes the form

$$\frac{d}{dx} \left[(1 - x^2) \frac{d\Theta}{dx} \right] = -\lambda \Theta.$$

This is a Sturm-Liouville equation with $p(x) = 1 - x^2$ and $q(x) = 0$.

Remark. Since $p(x) = 1 - x^2$ vanishes at the boundary points $x = \pm 1$, the boundary terms in the self-adjointness condition vanish automatically. The operator is self-adjoint provided solutions remain regular (finite) at $x = \pm 1$.

We seek power series solutions $\Theta(x) = \sum_{n=0}^{\infty} a_n x^n$. Substituting into Legendre's equation:

$$(1 - x^2) \sum_{n=0}^{\infty} a_n n(n-1) x^{n-2} - 2 \sum_{n=0}^{\infty} a_n n x^n + \lambda \sum_{n=0}^{\infty} a_n x^n = 0.$$

Matching coefficients yields the recurrence relation

$$a_{n+2} = \frac{n(n+1) - \lambda}{(n+2)(n+1)} a_n.$$

This relates even coefficients to even and odd to odd, giving two linearly independent solutions:

$$\begin{aligned} \Theta_0(x) &= a_0 \left[1 - \frac{\lambda}{2} x^2 - \frac{(6-\lambda)\lambda}{4!} x^4 + \dots \right] \quad (\text{even}) \\ \Theta_1(x) &= a_1 \left[x + \frac{(2-\lambda)}{3!} x^3 + \frac{(12-\lambda)(2-\lambda)}{5!} x^5 + \dots \right] \quad (\text{odd}). \end{aligned}$$

Proposition (Eigenvalue quantization). For solutions to remain regular at $x = \pm 1$, we must have $\lambda = \ell(\ell+1)$ for some $\ell \in \mathbb{N}_0$.

Proof. Since $\lim_{n \rightarrow \infty} \frac{a_{n+2}}{a_n} = 1$, the ratio test is inconclusive at $x = \pm 1$, and more refined tests show the infinite series diverges there. For regularity, the series must truncate. Setting $\lambda = \ell(\ell+1)$ makes $a_{\ell+2} = 0$, truncating to a polynomial. $\square$

Remark. This eigenvalue quantization is the mathematical origin of angular momentum quantization in quantum mechanics.

Definition (Legendre polynomials). The *Legendre polynomials* $P_\ell(x)$ for $\ell \in \mathbb{N}_0$ are the polynomial solutions to Legendre's equation with $\lambda = \ell(\ell + 1)$, normalized so that $P_\ell(1) = 1$. The first few are:

$$P_0(x) = 1, \quad P_1(x) = x, \quad P_2(x) = \frac{1}{2}(3x^2 - 1), \quad P_3(x) = \frac{1}{2}(5x^3 - 3x).$$

Proposition (Rodrigues formula). The Legendre polynomials satisfy

$$P_\ell(x) = \frac{1}{2^\ell \ell!} \frac{d^\ell}{dx^\ell} (x^2 - 1)^\ell.$$

Proof. One can verify this formula satisfies Legendre's equation. To check normalization:

$$P_\ell(1) = \frac{1}{2^\ell \ell!} \frac{d^\ell}{dx^\ell} [(x-1)^\ell (x+1)^\ell] \Big|_{x=1} = \frac{1}{2^\ell \ell!} [\ell! (x+1)^\ell + (x-1)(\cdots)]_{x=1} = 1 \square$$

Lemma. For $0 \leq k \leq \ell$, we have $\frac{d^k}{dx^k} (x^2 - 1)^\ell = Q_{\ell,k}(x)(x^2 - 1)^{\ell-k}$ for some polynomial $Q_{\ell,k}$ of degree k .

Proof. By induction on k . The base case $k = 0$ is trivial. For the inductive step:

$$\begin{aligned} \frac{d^{k+1}}{dx^{k+1}} (x^2 - 1)^\ell &= \frac{d}{dx} [Q_{\ell,k}(x)(x^2 - 1)^{\ell-k}] \\ &= (x^2 - 1)^{\ell-k-1} [Q'_{\ell,k}(x^2 - 1) + 2(\ell - k)Q_{\ell,k}x]. \end{aligned}$$

Since $\deg Q_{\ell,k} = k$, the bracketed expression has degree $k + 1$. $\square$

Theorem (Orthogonality of Legendre polynomials). The Legendre polynomials satisfy

$$(P_\ell, P_{\ell'}) = \int_{-1}^1 P_\ell(x) P_{\ell'}(x) dx = \frac{2\delta_{\ell\ell'}}{2\ell + 1}.$$

Proof. Using the Rodrigues formula:

$$(P_\ell, P_{\ell'}) = \frac{1}{2^\ell \ell!} \int_{-1}^1 \frac{d^\ell}{dx^\ell} (x^2 - 1)^\ell P_{\ell'}(x) dx.$$

Integrating by parts, the boundary terms vanish by Lemma 4.4.1 since each derivative of $(x^2 - 1)^\ell$ retains a factor of $(x^2 - 1)$ until the ℓ -th derivative. Thus we can transfer derivatives to $P_{\ell'}$.

If $\ell' < \ell$, integrating by parts ℓ' times gives the ℓ' -th derivative of $P_{\ell'}$, which is a constant times $\ell'!$. But we need ℓ derivatives total, and the remaining $(\ell - \ell')$ derivatives annihilate the constant. Hence $(P_\ell, P_{\ell'}) = 0$ for $\ell \neq \ell'$.

The normalization $2/(2\ell + 1)$ follows from direct calculation. $\square$

Proposition (Roots of Legendre polynomials). All ℓ roots of $P_\ell(x)$ are real and lie in $(-1, 1)$.

Proof. Suppose only $m < \ell$ roots lie in $(-1, 1)$. Let $Q_m(x) = \prod_{r=1}^m (x - x_r)$ where $\{x_1, \dots, x_m\}$ are these roots. Then $P_\ell(x)Q_m(x)$ does not change sign on $(-1, 1)$: the roots outside $(-1, 1)$ contribute no sign changes there, and the roots inside appear squared. Hence

$$\pm \int_{-1}^1 P_\ell(x)Q_m(x) \, dx > 0.$$

But expanding $Q_m(x) = \sum_{r=0}^m q_r P_r(x)$ and using orthogonality, $(P_\ell, Q_m) = 0$ since $m < \ell$. Contradiction. $\square$

4.4.2 Solution to radial part

With $\Theta(\theta) = P_\ell(\cos \theta)$, the radial equation becomes

$$(r^2 R')' = \ell(\ell + 1)R.$$

Proposition (Radial solutions). Trying $R(r) = r^\alpha$ gives $\alpha(\alpha + 1) = \ell(\ell + 1)$, so $\alpha = \ell$ or $\alpha = -(\ell + 1)$.

Theorem (General solution in spherical coordinates). The general axisymmetric solution to Laplace's equation is

$$\phi(r, \theta) = \sum_{\ell=0}^{\infty} \left(a_\ell r^\ell + \frac{b_\ell}{r^{\ell+1}} \right) P_\ell(\cos \theta).$$

For regularity at $r = 0$, we require $b_\ell = 0$ for all ℓ .

Example (Dirichlet problem on a ball). To solve $\nabla^2 \phi = 0$ on $\{r \leq a\}$ with $\phi(a, \theta) = f(\theta)$: expand f in Legendre polynomials,

$$f(\theta) = \sum_{\ell=0}^{\infty} F_\ell P_\ell(\cos \theta), \quad F_\ell = \frac{2\ell + 1}{2} \int_{-1}^1 P_\ell(x) f(\arccos x) \, dx.$$

Then the solution is

$$\phi(r, \theta) = \sum_{\ell=0}^{\infty} F_\ell \left(\frac{r}{a} \right)^\ell P_\ell(\cos \theta).$$

4.5 Multipole expansions for Laplace's equation

Proposition (Fundamental solution). The function $\phi(\mathbf{r}) = \frac{1}{|\mathbf{r}-\mathbf{r}'|}$ solves Laplace's equation $\nabla^2\phi = 0$ for all $\mathbf{r} \in \mathbb{R}^3 \setminus \{\mathbf{r}'\}$.

Theorem (Multipole expansion). For $|\mathbf{r}| < |\mathbf{r}'|$,

$$\frac{1}{|\mathbf{r}-\mathbf{r}'|} = \frac{1}{r'} \sum_{\ell=0}^{\infty} \left(\frac{r}{r'}\right)^{\ell} P_{\ell}(\hat{\mathbf{r}} \cdot \hat{\mathbf{r}}'),$$

where $r = |\mathbf{r}|$, $r' = |\mathbf{r}'|$, and $\hat{\mathbf{r}} = \mathbf{r}/r$.

Proof. For $\mathbf{r}' = \hat{\mathbf{k}}$ (unit vector in z direction), we have

$$\frac{1}{|\mathbf{r}-\hat{\mathbf{k}}|} = \frac{1}{\sqrt{r^2 + 1 - 2r \cos \theta}} = \sum_{\ell=0}^{\infty} c_{\ell} r^{\ell} P_{\ell}(\cos \theta).$$

At $\theta = 0$, using $P_{\ell}(1) = 1$:

$$\sum_{\ell=0}^{\infty} c_{\ell} r^{\ell} = \frac{1}{1-r} = \sum_{\ell=0}^{\infty} r^{\ell},$$

so $c_{\ell} = 1$ for all ℓ . The general case follows by rotational symmetry. $\square$

Definition (Multipole terminology). Expanding the multipole series:

$$\frac{1}{|\mathbf{r}-\mathbf{r}'|} = \underbrace{\frac{1}{r'}}_{\text{monopole}} + \underbrace{\frac{r}{r'^2} \hat{\mathbf{r}} \cdot \hat{\mathbf{r}}'}_{\text{dipole}} + \dots$$

In electromagnetism, the monopole term corresponds to a point charge, and the dipole term to an electric dipole.

4.6 Laplace's equation in cylindrical coordinates

Definition (Cylindrical coordinates). *Cylindrical coordinates* (r, θ, z) on $\mathbb{R}^3$ have Laplacian

$$\nabla^2\phi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2}.$$

Example (Laplace's equation on a half-cylinder). Let $\Omega = \{(r, \theta, z) \in \mathbb{R}^3 : r \leq a, z \geq 0\}$. We seek a regular solution to $\nabla^2\phi = 0$ with boundary

conditions:

$$\phi(a, \theta, z) = 0, \quad \phi(r, \theta, 0) = f(r, \theta), \quad \lim_{z \rightarrow \infty} \phi(r, \theta, z) = 0.$$

Step 1: Separation of variables. With $\phi(r, \theta, z) = R(r)\Theta(\theta)Z(z)$:

$$\frac{1}{rR}(rR')' + \frac{1}{r^2} \frac{\Theta''}{\Theta} + \frac{Z''}{Z} = 0.$$

The separated equations are $Z'' = \mu Z$, $\Theta'' = -\lambda \Theta$, and

$$r^2 R'' + rR' + (\mu r^2 - \lambda)R = 0.$$

Step 2: Angular and axial parts. Periodicity in θ requires

$$\Theta(\theta) = a_n \sin n\theta + b_n \cos n\theta, \quad \lambda = n^2, \quad n \in \mathbb{N}_0.$$

Decay as $z \rightarrow \infty$ requires $Z(z) = c_\mu e^{-\sqrt{\mu}z}$ (with $\mu > 0$).

Step 3: Radial equation. The radial equation is Sturm-Liouville type:

$$\frac{d}{dr} \left(r \frac{dR}{dr} \right) - \frac{n^2}{r} R = -\mu r R,$$

with $p(r) = r$, $q(r) = -n^2/r$, $w(r) = r$. Substituting $x = r\sqrt{\mu}$ gives Bessel's equation.

Definition (Bessel's equation and Bessel functions). *Bessel's equation of order n is*

$$x^2 \frac{d^2 R}{dx^2} + x \frac{dR}{dx} + (x^2 - n^2)R = 0.$$

Its two linearly independent solutions are *Bessel functions*:

- $J_n(x)$: Bessel function of the first kind, regular at the origin with $J_n(x) \sim x^n$ as $x \rightarrow 0$.
- $Y_n(x)$: Bessel function of the second kind, singular at the origin with $Y_0(x) \sim \ln x$ and $Y_n(x) \sim x^{-n}$ for $n > 0$.

Remark. Bessel functions resemble decaying sine waves, but their zeros are not regularly spaced.

Continuing Example 4.6:

Step 4: Apply regularity and boundary conditions. For regularity at $r = 0$, we discard Y_n terms. The separated solution is

$$\phi(r, \theta, z) = (a_n \sin n\theta + b_n \cos n\theta) e^{-\sqrt{\mu}z} J_n(r\sqrt{\mu}).$$

The boundary condition $\phi(a, \theta, z) = 0$ requires $J_n(a\sqrt{\mu}) = 0$, so

$$\sqrt{\mu} = \frac{k_{ni}}{a},$$

where k_{ni} denotes the i -th positive zero of J_n .

Step 5: General solution. The general solution satisfying homogeneous conditions is

$$\phi(r, \theta, z) = \sum_{n=0}^{\infty} \sum_{i=1}^{\infty} (A_{ni} \sin n\theta + B_{ni} \cos n\theta) \exp\left(-\frac{k_{ni}}{a} z\right) J_n\left(\frac{k_{ni}r}{a}\right).$$

Proposition (Orthogonality of Bessel functions). With weight function $w(r) = r$:

$$\int_0^a J_n\left(\frac{k_{nj}r}{a}\right) J_n\left(\frac{k_{ni}r}{a}\right) r \, dr = \frac{a^2}{2} \delta_{ij} [J'_n(k_{ni})]^2.$$

This relates zeros of Bessel functions of the *same order* n .

Step 6: Determine coefficients. Imposing $\phi(r, \theta, 0) = f(r, \theta)$ and using orthogonality in both θ and r :

$$B_{mj} = \frac{2}{\pi a^2 [J'_m(k_{mj})]^2} \int_0^a \int_{-\pi}^{\pi} \cos m\theta J_m\left(\frac{k_{mj}r}{a}\right) f(r, \theta) r \, dr \, d\theta.$$

Remark. In practice, these integrals are often computed numerically, since the zeros k_{mj} of Bessel functions have no simple closed form.

4.7 The heat equation

Definition (Heat equation). The *heat equation* on $\Omega \subseteq \mathbb{R}^n$ for a function $\phi : \Omega \times [0, \infty) \rightarrow \mathbb{R}$ is

$$\frac{\partial \phi}{\partial t} = \kappa \nabla^2 \phi,$$

where $\kappa > 0$ is the *diffusion constant*. Physically, ϕ represents temperature and the equation states that the rate of change of temperature is proportional to the Laplacian.

Remark. In equilibrium ($\partial\phi/\partial t = 0$), the heat equation reduces to Laplace's equation.

Proposition (Conservation of total heat). If ϕ solves the heat equation and $\nabla\phi \rightarrow 0$ as $|\mathbf{x}| \rightarrow \infty$, then

$$\frac{d}{dt} \int_{\mathbb{R}^n} \phi(\mathbf{x}, t) \, d^n x = 0.$$

Proof.

$$\frac{d}{dt} \int_{\mathbb{R}^n} \phi(\mathbf{x}, t) \, d^n x = \int_{\mathbb{R}^n} \frac{\partial \phi}{\partial t} \, d^n x = \kappa \int_{\mathbb{R}^n} \nabla^2 \phi \, d^n x = 0,$$

since the last integral vanishes by the divergence theorem. $\square$

Proposition (Symmetries of the heat equation). If $\phi(\mathbf{x}, t)$ solves the heat equation, so do:

- (i) Translation: $\phi_1(\mathbf{x}, t) = \phi(\mathbf{x} - \mathbf{x}_0, t - t_0)$
- (ii) Scaling: $\phi_2(\mathbf{x}, t) = A\phi(\lambda\mathbf{x}, \lambda^2 t)$ for any $A, \lambda \in \mathbb{R}$.

The scaling symmetry suggests seeking self-similar solutions. If $\phi_2 = \lambda^n \phi(\lambda\mathbf{x}, \lambda^2 t)$ preserves total heat, setting $\lambda = 1/\sqrt{\kappa t}$ suggests the ansatz

$$\phi(\mathbf{x}, t) = \frac{1}{(\kappa t)^{n/2}} F\left(\frac{\mathbf{x}}{\sqrt{\kappa t}}\right) = \frac{1}{(\kappa t)^{n/2}} F(\boldsymbol{\eta}), \quad \boldsymbol{\eta} = \frac{\mathbf{x}}{\sqrt{\kappa t}}.$$

Example (Self-similar solution in one dimension). In $1 + 1$ dimensions, we seek $\phi(x, t) = \frac{1}{\sqrt{\kappa t}} F(\eta)$ where $\eta = x/\sqrt{\kappa t}$.

Computing derivatives:

$$\frac{\partial \phi}{\partial t} = \frac{-1}{2\sqrt{\kappa t^3}} [F + \eta F'], \quad \kappa \frac{\partial^2 \phi}{\partial x^2} = \frac{1}{\sqrt{\kappa t^3}} F''.$$

The heat equation becomes $(2F' + \eta F)' = 0$, so $2F' + \eta F = C$. With decay at infinity, $C = 0$, giving $F' = -\frac{\eta}{2} F$, hence

$$F(\eta) = a \exp\left(-\frac{\eta^2}{4}\right).$$

Normalizing to $\int_{-\infty}^{\infty} \phi dx = 1$ gives $a = 1/\sqrt{4\pi}$.

Definition (Heat kernel). The *heat kernel* (or *fundamental solution*) on $\mathbb{R}^n$ is

$$K(\mathbf{x}, t; \mathbf{x}_0, t_0) = \frac{1}{(4\pi\kappa(t-t_0))^{n/2}} \exp\left(-\frac{|\mathbf{x} - \mathbf{x}_0|^2}{4\kappa(t-t_0)}\right), \quad t > t_0.$$

This is a Gaussian centered at $\mathbf{x}_0$ with standard deviation $\sqrt{2\kappa(t-t_0)}$. As t increases, the solution spreads out and flattens while preserving total integral.

Proposition (Positivity of Laplacian eigenvalues). On a closed compact domain Ω (or with Dirichlet boundary conditions), all eigenvalues of $-\nabla^2$ are non-negative.

Proof. Let $\nabla^2 y = -\lambda y$. Then

$$-\lambda \int_{\Omega} |y|^2 d^n x = \int_{\Omega} y^* \nabla^2 y d^n x = \int_{\partial\Omega} y^* \mathbf{n} \cdot \nabla y d^{n-1} x - \int_{\Omega} |\nabla y|^2 d^n x.$$

If the boundary term vanishes, then $\lambda = \frac{\int_{\Omega} |\nabla y|^2 d^n x}{\int_{\Omega} |y|^2 d^n x} \geq 0$. $\square$

Theorem (Solution by eigenfunction expansion). Let $\{y_I\}$ be eigenfunctions of $-\nabla^2$ with eigenvalues $\lambda_I \geq 0$. If $\phi(\mathbf{x}, 0) = \sum_I c_I y_I(\mathbf{x})$, then the solution to the heat equation (with $\kappa = 1$) is

$$\phi(\mathbf{x}, t) = \sum_I c_I e^{-\lambda_I t} y_I(\mathbf{x}).$$

Remark. Since all $\lambda_I > 0$, the coefficients $c_I(t) = c_I e^{-\lambda_I t}$ decay exponentially. Eigenfunctions with larger eigenvalues (corresponding to more oscillatory, “spiky” modes) decay fastest, explaining why heat flow smooths out solutions.

4.7.1 Brownian motion and the heat equation

The heat equation is not time-reversible, yet arises from microscopic dynamics that are. Einstein showed how this emerges from Brownian motion.

Example (Derivation from random walk). Let $p(y, \Delta t)$ be the probability of a particle moving by y in time Δt , with:

- Normalization: $\int_{-\infty}^{\infty} p(y, \Delta t) dy = 1$
- Symmetry: $p(y, \Delta t) = p(-y, \Delta t)$
- Localization: p is strongly peaked around $y = 0$

Let $P(x, t)$ be the probability density at position x , time t . Then

$$P(x, t + \Delta t) = \int_{-\infty}^{\infty} P(x - y, t) p(y, \Delta t) dy.$$

Taylor expanding $P(x - y, t)$ and using $\langle y \rangle = 0$ (by symmetry):

$$P(x, t + \Delta t) - P(x, t) \approx \frac{1}{2} \langle y^2 \rangle \frac{\partial^2 P}{\partial x^2},$$

where $\langle y^r \rangle = \int y^r p(y, \Delta t) dy$. If $\langle y^2 \rangle / (2\Delta t) \rightarrow \kappa$ as $\Delta t \rightarrow 0$, we obtain

$$\frac{\partial P}{\partial t} = \kappa \frac{\partial^2 P}{\partial x^2}.$$

Theorem (Uniqueness for the heat equation). Suppose $\phi : \Omega \times [0, \infty) \rightarrow \mathbb{R}$ satisfies the heat equation $\frac{\partial \phi}{\partial t} = \kappa \nabla^2 \phi$, with:

- Initial condition: $\phi(\mathbf{x}, 0) = f(\mathbf{x})$ for all $\mathbf{x} \in \Omega$
- Boundary condition: $\phi(\mathbf{x}, t)|_{\partial\Omega} = g(\mathbf{x}, t)$ for all $t \in [0, \infty)$.

Then ϕ is unique.

Proof. Suppose ϕ_1 and ϕ_2 are both solutions. Then define $\Phi = \phi_1 - \phi_2$ and

$$E(t) = \frac{1}{2} \int_{\Omega} \Phi^2 \, dV.$$

Then we know that $E(t) \geq 0$. Since ϕ_1, ϕ_2 both obey the heat equation, so does Φ . Also, on the boundary and at $t = 0$, we know that $\Phi = 0$. We have

$$\begin{aligned} \frac{dE}{dt} &= \int_{\Omega} \Phi \frac{d\Phi}{dt} \, dV \\ &= \kappa \int_{\Omega} \Phi \nabla^2 \Phi \, dV \\ &= \kappa \int_{\partial\Omega} \Phi \nabla \Phi \cdot d\mathbf{S} - \kappa \int_{\Omega} (\nabla \Phi)^2 \, dV \\ &= -\kappa \int_{\Omega} (\nabla \Phi)^2 \, dV \\ &\leq 0. \end{aligned}$$

So we know that E decreases with time but is always non-negative. We also know that at time $t = 0$, $E = \Phi = 0$. So $E = 0$ always. So $\Phi = 0$ always. So $\phi_1 = \phi_2$. $\square$

Example (Heat conduction in soil). Consider temperature $\phi(x, t)$ in soil at depth $x \geq 0$, where the surface temperature varies with daily and seasonal cycles.

We let $\phi(x, t)$ be the temperature of the soil as a function of the depth x , defined on $\mathbb{R}^{\geq 0} \times [0, \infty)$. Then it obeys the heat equation with conditions

$$(i) \quad \phi(0, t) = \phi_0 + A \cos\left(\frac{2\pi t}{t_D}\right) + B \cos\left(\frac{2\pi t}{t_Y}\right).$$

$$(ii) \quad \phi(x, t) \rightarrow \text{const as } x \rightarrow \infty.$$

We know that

$$\frac{\partial \phi}{\partial t} = K \frac{\partial^2 \phi}{\partial x^2}.$$

We try the separation of variables

$$\phi(x, t) = T(t)X(x).$$

Then we get the equations

$$T' = \lambda T, \quad X'' = \frac{\lambda}{K} X.$$

From the boundary solution, we know that our things will be oscillatory. So we let λ be imaginary, and set $\lambda = i\omega$. So we have

$$\phi(x, t) = e^{i\omega t} \left(a_{\omega} e^{-\sqrt{i\omega/K}x} + b_{\omega} e^{\sqrt{i\omega/K}x} \right).$$

Note that we have

$$\sqrt{\frac{i\omega}{K}} = \begin{cases} (1+i)\sqrt{\frac{|\omega|}{2K}} & \omega > 0 \\ (i-1)\sqrt{\frac{|\omega|}{2K}} & \omega < 0 \end{cases}$$

Since $\phi(x, t) \rightarrow \text{constant}$ as $x \rightarrow \infty$, we don't want our ϕ to blow up. So if $\omega < 0$, we need $a_\omega = 0$. Otherwise, we need $b_\omega = 0$.

To match up at $x = 0$, we just want terms with

$$|\omega| = \omega_D = \frac{2\pi}{t_D}, \quad |\omega| = \omega_Y = \frac{2\pi}{t_Y}.$$

So we can write out solution as

$$\begin{aligned} \phi(x, t) = & \phi_0 + A \exp\left(-\sqrt{\frac{\omega_D}{2K}}\right) \cos\left(\omega_D t - \sqrt{\frac{\omega_D}{2K}}x\right) \\ & + B \exp\left(-\sqrt{\frac{\omega_Y}{2K}}\right) \cos\left(\omega_Y t - \sqrt{\frac{\omega_Y}{2K}}x\right) \end{aligned}$$

We can notice a few things. Firstly, as we go further down, the effect of the sun decays, and the temperature is more stable. Also, the effect of the day-night cycle decays more quickly than the annual cycle, which makes sense. We also see that while the temperature does fluctuate with the same frequency as the day-night/annual cycle, as we go down, there is a phase shift. This is helpful since we can store things underground and make them cool in summer, warm in winter.

Example (Heat conduction in a finite rod). Consider a rod of length $2L$ from $x = -L$ to $x = L$.

$$\begin{array}{ccc} \bullet & \bullet & \bullet \\ x = -L & x = 0 & x = L \end{array}$$

We have the initial conditions

$$\phi(x, 0) = \Theta(x) = \begin{cases} 1 & 0 < x < L \\ 0 & -L < x < 0 \end{cases}$$

and the boundary conditions

$$\phi(-L, t) = 0, \quad \phi(L, t) = 1$$

So we start with a step temperature, and then maintain the two ends at fixed temperatures 0 and 1.

We are going to do separation of variables, but we note that all our boundary and initial conditions are inhomogeneous. This is not helpful. So we use a little trick. We first look for *any* solution satisfying the boundary conditions $\phi_S(-L, t) = 0$, $\phi_S(L, t) = 1$. For example, we can look for time-independent solutions $\phi_S(x, t) = \phi_S(x)$. Then we need $\frac{d^2 \phi_S}{dx^2} = 0$. So we get

$$\phi_S(x) = \frac{x + L}{2L}.$$

By linearity, $\psi(x, t) = \phi(x, t) - \phi_S(x)$ obeys the heat equation with the conditions

$$\psi(-L, t) = \psi(L, t) = 0,$$

which is homogeneous! Our initial condition now becomes

$$\psi(x, 0) = \Theta(x) - \frac{x + L}{2L}.$$

We now perform separation of variables for

$$\psi(x, t) = X(x)T(t).$$

Then we obtain the equations

$$T' = -\kappa\lambda T, \quad X' = -\lambda X.$$

Then we have

$$\psi(x, t) = \left[a \sin(\sqrt{\lambda}x) + b \cos(\sqrt{\lambda}x) \right] e^{-\kappa\lambda t}.$$

Since initial condition is odd, we can eliminate all cos terms. Our boundary conditions also requires

$$\lambda = \frac{n^2\pi^2}{L^2}, \quad n = 1, 2, \dots$$

So we have

$$\phi(x, t) = \phi_s(x) + \sum_{n=1}^{\infty} a_n \sin\left(\frac{n\pi x}{L}\right) \exp\left(-\frac{\kappa n^2\pi^2}{L^2}t\right),$$

where a_n are the Fourier coefficients

$$a_n = \frac{1}{L} \int_{-L}^L \left[\Theta(x) - \frac{x + L}{2L} \right] \sin\left(\frac{n\pi x}{L}\right) dx = \frac{1}{n\pi}.$$

Example (Cooling of a uniform sphere). Once upon a time, Charles Darwin went around the Earth, looked at species, and decided that evolution happened. When he came up with his theory, it worked quite well, except that there was one worry. Was there enough time on Earth for evolution to occur?

This calculation was done by Kelvin. He knew well that the Earth started as a ball of molten rock, and obviously life couldn't have evolved when the world was still molten rock. So he would want to know how long it took for the Earth to cool down to its current temperature, and if that was sufficient for evolution to occur.

We can, unsurprisingly, use the heat equation. We assume that at time $t = 0$, the temperature is ϕ_0 , the melting temperature of rock. We also assume that the space is cold, and we let the temperature on the boundary of Earth as 0. We then solve the heat equation on a sphere (or ball), and see how much time it takes for Earth to cool down to its present temperature.

We let $\Omega = \{(x, y, z) \in \mathbb{R}^3, r \leq R\}$, and we want a solution $\phi(r, t)$ of the heat equation that is spherically symmetric and obeys the conditions

- $\phi(R, t) = 0$
- $\phi(r, 0) = \phi_0$,

We can write the heat equation as

$$\frac{\partial \phi}{\partial t} = \kappa \nabla^2 \phi = \frac{\kappa}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \phi}{\partial r} \right).$$

Again, we do the separation of variables.

$$\phi(r, t) = R(r)T(t).$$

So we get

$$T' = -\lambda^2 \kappa T, \quad \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) = -\lambda^2 r^2 R.$$

We can simplify this a bit by letting $R(r) = \frac{S(r)}{r}$, then our radial equation just becomes

$$S'' = -\lambda^2 S.$$

We can solve this to get

$$R(r) = A_\lambda \frac{\sin \lambda r}{r} + B_\lambda \frac{\cos \lambda r}{r}.$$

We want a regular solution at $r = 0$. So we need $B_\lambda = 0$. Also, the boundary condition $\phi(R, t) = 0$ gives

$$\lambda = \frac{n\pi}{R}, \quad n = 1, 2, \dots$$

So we get

$$\phi(r, t) = \sum_{n \in \mathbb{Z}} \frac{A_n}{r} \sin \left(\frac{n\pi r}{R} \right) \exp \left(\frac{-\kappa n^2 \pi^2 t}{R^2} \right),$$

where our coefficients are

$$A_n = (-1)^{n+1} \frac{\phi_0 R}{n\pi}.$$

We know that the Earth isn't just a cold piece of rock. There are still volcanoes. So we know many terms still contribute to the sum nowadays. This is rather difficult to work with. So we instead look at the temperature gradient

$$\frac{\partial \phi}{\partial r} = \frac{\phi_0}{r} \sum_{n \in \mathbb{Z}} (-1)^{n+1} \cos \left(\frac{n\pi r}{R} \right) \exp \left(\frac{-\kappa n^2 \pi^2 t}{R^2} \right) + \sin \text{ stuff}.$$

We evaluate this at the surface of the Earth, $R = r$. So we get the gradient

$$\left. \frac{\partial \phi}{\partial r} \right|_R = -\frac{\phi_0}{R} \sum_{n \in \mathbb{Z}} \exp\left(-\frac{\kappa n^2 \pi^2 t}{R^2}\right) \approx \frac{\phi_0}{R} \int_{-\infty}^{\infty} \exp\left(-\frac{\kappa \pi y^2 t}{R^2}\right) dy = \phi_0 \sqrt{\frac{1}{\pi \kappa t}}.$$

So the age of the Earth is approximately

$$t \approx \left(\frac{\phi_0}{V}\right)^2 \frac{1}{4\pi K},$$

where

$$V = \left. \frac{\partial \phi}{\partial r} \right|_R.$$

We can plug the numbers in, and get that the Earth is 100 million years. This is not enough time for evolution.

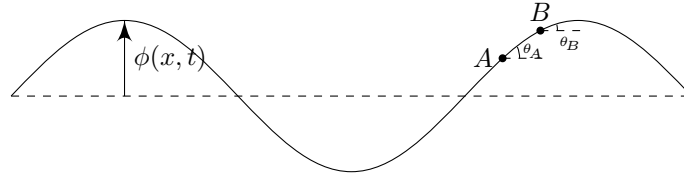
Later on, people discovered that fission of metals inside the core of Earth produce heat, and provides an alternative source of heat. So problem solved! The current estimate of the age of the universe is around 4 billion years, and evolution did have enough time to occur.

4.8 The wave equation

We derive the wave equation from physical principles and study its solutions.

4.8.1 Derivation from string dynamics

Consider a string $x \in [0, L]$ undergoing small transverse oscillations described by displacement $\phi(x, t)$.



Consider two points A, B separated by a small distance δx . Let T_A (T_B) be the outward pointing tension tangent to the string at A (B). Since there is no sideways (x) motion, there is no net horizontal force. So

$$T_A \cos \theta_A = T_B \cos \theta_B = T. \quad (*)$$

If the string has mass per unit length μ , then in the vertical direction, Newton's second law gives

$$\mu \delta x \frac{\partial^2 \phi}{\partial t^2} = T_B \sin \theta_B - T_A \sin \theta_A.$$

We now divide everything by T , noting the relation (*), and get

$$\begin{aligned}\mu \frac{\delta x}{T} \frac{\partial^2 \phi}{\partial t^2} &= \frac{T_B \sin \theta_B}{T_B \cos \theta_B} - \frac{T_A \sin \theta_A}{T_A \cos \theta_A} \\ &= \tan \theta_B - \tan \theta_A \\ &= \frac{\partial \phi}{\partial x} \Big|_B - \frac{\partial \phi}{\partial x} \Big|_A \\ &\approx \delta x \frac{\partial^2 \phi}{\partial x^2}.\end{aligned}$$

Taking the limit $\delta x \rightarrow 0$ and setting $c^2 = T/\mu$, we obtain:

Definition (Wave equation). The *wave equation* in one spatial dimension is

$$\frac{\partial^2 \phi}{\partial t^2} = c^2 \frac{\partial^2 \phi}{\partial x^2},$$

where $c > 0$ is the *wave speed*. For a string with tension T and mass per unit length μ , we have $c = \sqrt{T/\mu}$.

Remark. The general solution in one dimension is $\phi(x, t) = f(x - ct) + g(x + ct)$, representing left- and right-traveling waves. However, separation of variables generalizes to higher dimensions.

Example (String with fixed ends). For a string fixed at both ends ($\phi(0, t) = \phi(L, t) = 0$), separation of variables gives

$$\phi(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi x}{L}\right) \left[A_n \cos\left(\frac{n\pi ct}{L}\right) + B_n \sin\left(\frac{n\pi ct}{L}\right) \right].$$

The coefficients A_n are determined by the initial displacement $\phi(x, 0)$, and B_n by the initial velocity $\frac{\partial \phi}{\partial t}(x, 0)$. Since the wave equation is second-order in time, *two* initial conditions are required.

4.8.2 Energetics and uniqueness

Definition (Energy of a vibrating string). For a vibrating string with displacement $\phi(x, t)$, mass density μ , and wave speed c :

- *Kinetic energy*: $K(t) = \frac{\mu}{2} \int_0^L \left(\frac{\partial \phi}{\partial t} \right)^2 dx$
- *Potential energy*: $V(t) = \frac{\mu c^2}{2} \int_0^L \left(\frac{\partial \phi}{\partial x} \right)^2 dx$
- *Total energy*: $E(t) = K(t) + V(t)$

The potential energy arises from the work done against tension due to string extension:

$$\begin{aligned} T \times \text{extension} &= T(\delta s - \delta x) = T(\sqrt{\delta x^2 + \delta \phi^2} - \delta x) \\ &\approx \frac{T}{2} \delta x \left(\frac{\partial \phi}{\partial x} \right)^2 = \frac{\mu c^2}{2} \delta x \left(\frac{\partial \phi}{\partial x} \right)^2. \end{aligned}$$

Using our series expansion for ϕ , we get

$$\begin{aligned} K(t) &= \frac{\mu \pi^2 c^2}{4L} \sum_{n=1}^{\infty} n^2 \left[A_n \sin \left(\frac{n\pi ct}{L} \right) - B_n \cos \left(\frac{n\pi ct}{L} \right) \right]^2 \\ V(t) &= \frac{\mu \pi^2 c^2}{4L} \sum_{n=1}^{\infty} n^2 \left[A_n \cos \left(\frac{n\pi ct}{L} \right) + B_n \sin \left(\frac{n\pi ct}{L} \right) \right]^2 \end{aligned}$$

The *total energy* is

$$E(t) = \frac{n\pi^2 c^2}{4L} \sum_{n=1}^{\infty} n^2 (A_n^2 + B_n^2).$$

What can we see here? Our solution is essentially an (infinite) sum of independent harmonic oscillators, one for each n . The period of the fundamental mode ($n = 1$) is $\frac{2\pi}{\omega} = 2\pi \cdot \frac{L}{\pi c} = \frac{2L}{c}$. Thus, averaging over a period, the average kinetic energy is

$$\bar{K} = \frac{c}{2L} \int_0^{2L/c} K(t) dt = \bar{V} = \frac{c}{2L} \int_0^{2L/c} V(t) dt = \frac{E}{2}.$$

Hence we have an equipartition of the energy between the kinetic and potential energy.

The energy also allows us to prove a uniqueness theorem. First we show that energy is conserved in general.

Theorem (Energy conservation for wave equation). If ϕ satisfies the wave equation $\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi$ on $\Omega \times (0, \infty)$ with $\phi|_{\partial\Omega}$ fixed, then the total energy $E = K + V$ is constant.

Proof. By definition, we have

$$\frac{dE}{dt} = \int_{\Omega} \frac{\partial^2 \phi}{\partial t^2} \frac{\partial \phi}{\partial t} + c^2 \nabla \left(\frac{\partial \phi}{\partial t} \right) \cdot \nabla \phi dV.$$

We integrate by parts in the second term to obtain

$$\frac{dE}{dt} = \int_{\Omega} \frac{d\phi}{dt} \left(\frac{\partial^2 \phi}{\partial t^2} - c^2 \nabla^2 \phi \right) dV + c^2 \int_{\partial\Omega} \frac{\partial \phi}{\partial t} \nabla \phi \cdot dS.$$

Since $\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi$ by the wave equation, and ϕ is constant on $\partial\Omega$, we know that

$$\frac{dE}{dt} = 0. \quad \square$$

Theorem (Uniqueness for wave equation). Suppose $\phi : \Omega \times [0, \infty) \rightarrow \mathbb{R}$ satisfies the wave equation $\frac{\partial^2 \phi}{\partial t^2} = c^2 \nabla^2 \phi$ with:

- (i) Initial displacement: $\phi(\mathbf{x}, 0) = f(\mathbf{x})$
- (ii) Initial velocity: $\frac{\partial \phi}{\partial t}(\mathbf{x}, 0) = g(\mathbf{x})$
- (iii) Boundary condition: $\phi|_{\partial\Omega \times [0, \infty)} = h(\mathbf{x}, t)$

Then ϕ is unique.

Proof. Suppose ϕ_1 and ϕ_2 are two such solutions. Then $\psi = \phi_1 - \phi_2$ obeys the wave equation

$$\frac{\partial^2 \psi}{\partial t^2} = c^2 \nabla^2 \psi,$$

and

$$\psi|_{\partial\Omega \times [0, \infty)} = \psi|_{\Omega \times \{0\}} = \frac{\partial \psi}{\partial t} \Big|_{\Omega \times \{0\}} = 0.$$

We let

$$E_\psi(t) = \frac{1}{2} \int_{\Omega} \left[\left(\frac{\partial \psi}{\partial t} \right)^2 + c^2 \nabla \psi \cdot \nabla \psi \right] dV.$$

Then since ψ obeys the wave equation with fixed boundary conditions, we know E_ψ is constant.

Initially, at $t = 0$, we know that $\psi = \frac{\partial \psi}{\partial t} = 0$. So $E_\psi(0) = 0$. At time t , we have

$$E_\psi = \frac{1}{2} \int_{\Omega} \left(\frac{\partial \psi}{\partial t} \right)^2 + c^2 (\nabla \psi) \cdot (\nabla \psi) dV = 0.$$

Hence we must have $\frac{\partial \psi}{\partial t} = 0$. So ψ is constant. Since it is 0 at the beginning, it is always 0. $\square$

Example (Circular drum). Consider a circular membrane $\Omega = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 1\}$. Let $\phi(r, \theta, t)$ solve

$$\frac{1}{c^2} \frac{\partial^2 \phi}{\partial t^2} = \nabla^2 \phi = \frac{1}{r} \left(r \frac{\partial \phi}{\partial r} \right)' + \frac{1}{r^2} \frac{\partial^2 \phi}{\partial \theta^2},$$

with the boundary condition $\phi|_{\partial\Omega} = 0$. We can imagine this as a drum, where the membrane can freely oscillate with the boundary fixed.

Separating variables with $\phi(r, \theta, t) = T(t)R(r)\Theta(\theta)$, we get

$$T'' = -c^2 \lambda T, \quad \Theta'' = -\mu \Theta, \quad r(R')' + (r^2 \lambda - \mu)R = 0.$$

Then as before, T and Θ are both sine and cosine waves. Since we are in polars coordinates, we need $\phi(t, r, \theta + 2\pi) = \phi(t, r, \theta)$. So we must have $\mu = m^2$ for some $m \in \mathbb{N}$. Then the radial equation becomes

$$r(rR')' + (r^2 \lambda - m^2)R = 0,$$

which is Bessel's equation of order m . So we have

$$R(r) = a_m J_m(\sqrt{\lambda} r) + b_m Y_m(\sqrt{\lambda} r).$$

Since we want regularity at $r = 0$, we need $b_m = 0$ for all m . To satisfy the boundary condition $\phi|_{\partial\Omega} = 0$, we must choose $\sqrt{\lambda} = k_{mi}$, where k_{mi} is the i th root of J_m .

Hence the general solution is

$$\begin{aligned}\phi(t, r, \theta) = & \sum_{i=0}^{\infty} [A_{0i} \sin(k_{0i}ct) + B_{0i} \cos(k_{0i}ct)] J_0(k_{0i}r) \\ & + \sum_{m=1}^{\infty} \sum_{i=0}^{\infty} [A_{mi} \cos(m\theta) + B_{mi} \sin(m\theta)] \sin k_{mi}ct J_m(k_{mi}r) \\ & + \sum_{m=1}^{\infty} \sum_{i=0}^{\infty} [C_{mi} \cos(m\theta) + D_{mi} \sin(m\theta)] \cos k_{mi}ct J_m(k_{mi}r)\end{aligned}$$

For example, suppose we have the initial conditions $\phi(0, r, \theta) = 0$, $\partial_t \phi(0, r, \theta) = g(r)$. So we start with a flat surface and suddenly hit it with some force. By symmetry, we must have $A_{mi}, B_{mi}, C_{mi}, D_{mi} = 0$ for $m \neq 0$. If this does not seem obvious, we can perform some integrals with the orthogonality relations to prove this.

At $t = 0$, we need $\phi = 0$. So we must have $B_{0i} = 0$. So we are left with

$$\phi = \sum_{i=0}^{\infty} A_{0i} \sin(k_{0i}ct) J_0(k_{0i}r).$$

We can differentiate this, multiply with $J_0(k_{0j}r)r$ to obtain

$$\int_0^1 \sum_{i=0}^{\infty} k_{0i} c A_{0i} J_0(k_{0i}r) J_0(k_{0j}r) r \, dr = \int_0^1 g(r) J_0(k_{0j}r) r \, dr.$$

Using the orthogonality relations for J_0 from the example sheet, we get

$$A_{0i} = \frac{2}{ck_{0i}} \frac{1}{[J'_0(k_{0i})]^2} \int_0^1 g(r) J_0(k_{0j}r) r \, dr.$$

Note that the frequencies come from the roots of the Bessel's function, and are not evenly spaced. This is different from, say, string instruments, where the frequencies are evenly spaced. So drums sound differently from strings.

So far, we have used separation of variables to solve our differential equations. This is quite good, as it worked in our examples, but there are a few issues with it. Of course, we have the problem of whether it converges or not, but there is a more fundamental problem.

To perform separation of variables, we need to pick a good coordinate system, such that the boundary conditions come in a nice form. We were able to perform separation of variables because we can find some coordinates that fit our boundary conditions well. However, in real life, most things are not nice. Our domain might have a weird shape, and we cannot easily find good coordinates for it.

Hence, instead of solving things directly, we want to ask some more general questions about them. In particular, Kac asked the question “can you hear the shape of a drum?” — suppose we know all the frequencies of the modes of oscillation on some domain Ω . Can we know what Ω is like?

The answer is no, and we can explicitly construct two distinct drums that sound the same. Fortunately, we get an affirmative answer if we restrict ourselves a bit. If we require Ω to be convex, and has a real analytic boundary, then yes! In fact, we have the following result: let $N(\lambda_0)$ be the number of eigenvalues less than λ_0 . Then we can show that

$$4\pi^2 \lim_{\lambda_0 \rightarrow \infty} \frac{N(\lambda_0)}{\lambda_0} = \text{Area}(\Omega).$$

5 Distributions

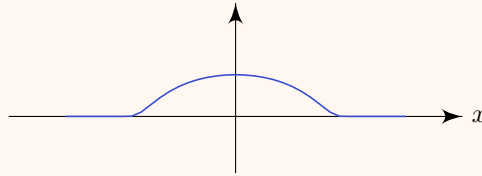
5.1 Distributions

When using separation of variables, we encounter convergence questions: does the infinite sum of separated solutions converge, and does it satisfy the differential equation? Since infinite sums of continuous functions can be discontinuous (as seen in Fourier series), we need a broader framework. This leads to *distributions*, or generalized functions.

Definition (Test functions). The space of *test functions* on $\Omega \subseteq \mathbb{R}^n$, denoted $D(\Omega)$ or $C_{\text{cpt}}^\infty(\Omega)$, consists of all infinitely differentiable functions with compact support in Ω .

Example (Bump function). A standard example is the bump function

$$\phi(x) = \begin{cases} e^{-1/(1-x^2)} & |x| < 1 \\ 0 & \text{otherwise.} \end{cases}$$



Definition (Distribution). A *distribution* on Ω is a continuous linear functional $T : D(\Omega) \rightarrow \mathbb{R}$. For $\phi \in D(\Omega)$, we write the value of T at ϕ as $T[\phi]$.

Example (Regular distribution). Any locally integrable function f defines a distribution T_f by

$$T_f[\phi] = \int_{\Omega} f(x)\phi(x) \, dx.$$

Linearity follows from linearity of integration. Thus every function can be viewed as a distribution.

The power of distributions lies in allowing “generalized functions” that are not ordinary functions.

Definition (Dirac delta distribution). The *Dirac delta* is the distribution $\delta : D(\Omega) \rightarrow \mathbb{R}$ defined by

$$\delta[\phi] = \phi(0).$$

More generally, for $a \in \Omega$, the *shifted Dirac delta* is $\delta_a[\phi] = \phi(a)$.

Remark. By analogy with regular distributions, we write symbolically

$$\delta[\phi] = \int_{\Omega} \delta(x)\phi(x) \, dx,$$

treating $\delta(x)$ as if it were a function. However, δ is *not* a function: if $\delta(x) = 0$ for $x \neq 0$ (as required since $\delta[\phi]$ depends only on $\phi(0)$), then the integral would vanish for any finite $\delta(0)$. Informally, one thinks of δ as “zero everywhere except at the origin, where it is infinite.”

Definition (Derivative of a distribution). The *derivative* of a distribution T is the distribution T' defined by

$$T'[\phi] = -T[\phi'].$$

More generally, the k -th derivative is $T^{(k)}[\phi] = (-1)^k T[\phi^{(k)}]$.

Remark. This definition is motivated by integration by parts: for a regular distribution T_f ,

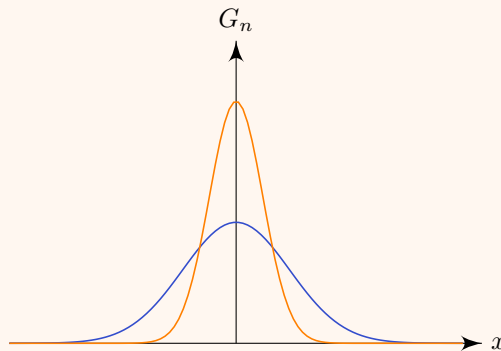
$$\int_{\Omega} f'(x)\phi(x) \, dx = - \int_{\Omega} f(x)\phi'(x) \, dx,$$

with no boundary terms since ϕ has compact support. Since test functions are infinitely differentiable, *every distribution has derivatives of all orders.*

Example (Delta as a limit of smooth functions). The Gaussians

$$G_n(x) = \frac{n}{\sqrt{\pi}} \exp(-n^2 x^2)$$

are smooth and satisfy $\int G_n = 1$. As $n \rightarrow \infty$, they become increasingly peaked at the origin:



As distributions, $G_n \rightarrow \delta$: for any test function ϕ , we have $\int G_n(x)\phi(x) \, dx \rightarrow \phi(0)$.

Example (Derivative of delta). By definition,

$$\delta'[\phi] = -\delta[\phi'] = -\phi'(0).$$

This can also be viewed as $\lim_{n \rightarrow \infty} G'_n$ in the distributional sense.

Proposition (Properties of the Dirac delta). The Dirac delta satisfies:

- (i) *Translation*: $\int_{-\infty}^{\infty} \delta(x - a)\phi(x) dx = \phi(a).$
- (ii) *Scaling*: For $c \neq 0$, $\int_{-\infty}^{\infty} \delta(cx)\phi(x) dx = \frac{1}{|c|}\phi(0)$, i.e., $\delta(cx) = \frac{1}{|c|}\delta(x).$
- (iii) *Composition*: If f is continuously differentiable with simple zeros at $\{x_i\}$, then

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x_i)|}.$$

Proof. For (1), substitute $y = x - a$. For (2), substitute $y = cx$ to get

$$\int \delta(cx)\phi(x) dx = \int \delta(y)\phi(y/c) \frac{dy}{|c|} = \frac{\phi(0)}{|c|}.$$

For (3), near each zero x_i , we have $f(x) \approx (x - x_i)f'(x_i)$, so by (2),

$$\int \delta(f(x))\phi(x) dx = \sum_i \int \delta((x - x_i)f'(x_i))\phi(x) dx = \sum_i \frac{\phi(x_i)}{|f'(x_i)|}. \quad \square$$

5.1.1 Eigenfunction expansions of delta

Proposition (Fourier expansion of delta). On the interval $[-L, L]$,

$$\delta(x) = \frac{1}{2L} \sum_{n \in \mathbb{Z}} e^{in\pi x/L}$$

in the sense of distributions.

Proof. Computing the Fourier coefficients:

$$\hat{\delta}_n = \frac{1}{2L} \int_{-L}^L e^{-in\pi x/L} \delta(x) dx = \frac{1}{2L}.$$

To verify this makes sense as a distribution, let $S_N \delta(x) = \frac{1}{2L} \sum_{n=-N}^N e^{in\pi x/L}$. Then

$$\begin{aligned} \lim_{N \rightarrow \infty} \int_{-L}^L S_N \delta(x) \phi(x) dx &= \lim_{N \rightarrow \infty} \sum_{n=-N}^N \frac{1}{2L} \int_{-L}^L e^{in\pi x/L} \phi(x) dx \\ &= \lim_{N \rightarrow \infty} \sum_{n=-N}^N \hat{\phi}_{-n} = \phi(0), \end{aligned}$$

since the Fourier series of the smooth function ϕ converges pointwise. $\square$

Proposition (General eigenfunction expansion of delta). Let $\{y_n(x)\}$ be a complete orthonormal set of eigenfunctions on $[a, b]$ with respect to weight $w(x)$, i.e.,

$$\int_a^b y_m^*(x) y_n(x) w(x) dx = \delta_{mn}.$$

Then

$$\delta(x - \xi) = w(\xi) \sum_n y_n^*(\xi) y_n(x) = w(x) \sum_n y_n^*(x) y_n(\xi).$$

Proof. Expanding $\delta(x - \xi) = \sum_n c_n y_n(x)$, the coefficients are

$$c_n = \int_a^b y_n^*(x) \delta(x - \xi) w(x) dx = y_n^*(\xi) w(\xi).$$

The two forms are equivalent since $\delta(x - \xi) = \frac{w(\xi)}{w(x)} \delta(x - \xi)$ (the factor only matters at $x = \xi$). $\square$

5.2 Green's functions

One of the main applications of the Dirac delta is in solving differential equations via Green's functions.

Definition (Green's function). Consider a linear differential operator $\mathcal{L}$ on $[a, b]$, such as

$$\mathcal{L} = \alpha(x) \frac{d^2}{dx^2} + \beta(x) \frac{d}{dx} + \gamma(x).$$

The *Green's function* $G(x, \xi)$ for $\mathcal{L}$ with given boundary conditions is the solution to

$$\mathcal{L}_x G(x, \xi) = \delta(x - \xi),$$

where $\mathcal{L}_x$ acts on the x variable, satisfying the same homogeneous boundary conditions.

Proposition (Solution via Green's function). If $G(x, \xi)$ is the Green's function for $\mathcal{L}$ with homogeneous boundary conditions, then the solution to $\mathcal{L}y = f$ with those boundary conditions is

$$y(x) = \int_a^b G(x, \xi) f(\xi) d\xi.$$

Proof. Apply $\mathcal{L}$ to the integral:

$$\mathcal{L}y = \int_a^b \mathcal{L}_x G(x, \xi) f(\xi) d\xi = \int_a^b \delta(x - \xi) f(\xi) d\xi = f(x). \quad \square$$

Remark. The Green's function represents the “inverse” of $\mathcal{L}$: symbolically, $y = \mathcal{L}^{-1}f$, where $\mathcal{L}^{-1}$ is an integral operator with kernel G .

5.2.1 Construction of Green's function for boundary value problems

We construct the Green's function for Dirichlet boundary conditions $G(a, \xi) = G(b, \xi) = 0$.

- (i) **Homogeneous solutions away from ξ :** Since $\mathcal{L}G = 0$ for $x \neq \xi$, we expand G in homogeneous solutions. Let y_1, y_2 be a basis for $\mathcal{L}y = 0$ with $y_1(a) = 0$ and $y_2(b) = 0$. Then

$$G(x, \xi) = \begin{cases} A(\xi)y_1(x) & a \leq x < \xi \\ B(\xi)y_2(x) & \xi < x \leq b. \end{cases}$$

- (ii) **Continuity condition:** If G were discontinuous at ξ , then G'' would contain δ' , which cannot be balanced by $\mathcal{L}G = \delta$. Hence G must be continuous:

$$A(\xi)y_1(\xi) = B(\xi)y_2(\xi). \quad (*)$$

- (iii) **Jump condition on derivative:** Integrating $\mathcal{L}G = \delta(x - \xi)$ over $(\xi - \varepsilon, \xi + \varepsilon)$:

$$\int_{\xi-\varepsilon}^{\xi+\varepsilon} [\alpha G'' + \beta G' + \gamma G] dx = 1.$$

The γG and $\beta G'$ terms vanish as $\varepsilon \rightarrow 0$ (since G is continuous and G' is bounded). Integration by parts gives

$$\alpha(\xi) (G'(\xi^+) - G'(\xi^-)) = 1,$$

yielding $B(\xi)y_2'(\xi) - A(\xi)y_1'(\xi) = 1/\alpha(\xi)$.

Definition (Wronskian). The *Wronskian* of two functions y_1, y_2 is $W = y_1y_2' - y_2y_1'$.

Theorem (Green's function for second-order BVP). For the operator $\mathcal{L} = \alpha(x)\frac{d^2}{dx^2} + \beta(x)\frac{d}{dx} + \gamma(x)$ with Dirichlet conditions on $[a, b]$, the Green's function is

$$G(x, \xi) = \frac{1}{\alpha(\xi)W(\xi)} \begin{cases} y_2(\xi)y_1(x) & a \leq x \leq \xi \\ y_1(\xi)y_2(x) & \xi < x \leq b, \end{cases}$$

where y_1, y_2 satisfy $\mathcal{L}y = 0$ with $y_1(a) = 0$, $y_2(b) = 0$, and W is their Wronskian.

Equivalently, using the Heaviside step function Θ :

$$G(x, \xi) = \frac{\Theta(\xi - x)y_2(\xi)y_1(x) + \Theta(x - \xi)y_1(\xi)y_2(x)}{\alpha(\xi)W(\xi)}.$$

The general solution to $\mathcal{L}y = f$ is then

$$y(x) = y_1(x) \int_x^b \frac{y_2(\xi)f(\xi)}{\alpha(\xi)W(\xi)} d\xi + y_2(x) \int_a^x \frac{y_1(\xi)f(\xi)}{\alpha(\xi)W(\xi)} d\xi.$$

Example (Green's function for $-y'' - y = f$). Consider $\mathcal{L}y = -y'' - y = f$ on $(0, 1)$ with $y(0) = y(1) = 0$.

The homogeneous equation $y'' + y = 0$ has solutions $\sin x$ and $\sin(1 - x)$ satisfying the boundary conditions at $x = 0$ and $x = 1$ respectively. The Wronskian is

$$W = -\sin x \cos(1 - x) - \sin(1 - x) \cos x = -\sin 1.$$

With $\alpha = -1$, the Green's function is

$$G(x, \xi) = \frac{\Theta(\xi - x) \sin(1 - \xi) \sin x + \Theta(x - \xi) \sin \xi \sin(1 - x)}{\sin 1}.$$

The solution is

$$y(x) = \frac{\sin x}{\sin 1} \int_x^1 \sin(1 - \xi) f(\xi) d\xi + \frac{\sin(1 - x)}{\sin 1} \int_0^x \sin \xi f(\xi) d\xi.$$

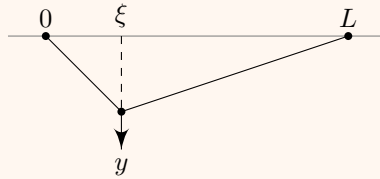
Example (String under gravity). A string with mass density $\mu(x)$ under tension T satisfies (from Newton's second law)

$$\mu \frac{\partial^2 y}{\partial t^2} = T \frac{\partial^2 y}{\partial x^2} + \mu g.$$

For the steady state ($\partial_t y = 0$) with $y(0) = y(L) = 0$:

$$\frac{d^2 y}{dx^2} = -\frac{\mu(x)g}{T}.$$

The Green's function satisfies $G'' = -\delta(x - \xi)$, representing a point mass at $x = \xi$:



The homogeneous solution $y'' = 0$ gives $y = Ax$ or $y = B(x - L)$. Thus

$$G(x, \xi) = \begin{cases} A(\xi)x & 0 \leq x < \xi \\ B(\xi)(x - L) & \xi < x \leq L. \end{cases}$$

Continuity ($A\xi = B(\xi - L)$) and the jump condition ($B - A = 1$, since $\alpha = -1$) give

$$A(\xi) = \frac{\xi - L}{L}, \quad B(\xi) = -\frac{\xi}{L}.$$

The Green's function is

$$G(x, \xi) = \frac{(\xi - L)x}{L} \Theta(\xi - x) - \frac{\xi(x - L)}{L} \Theta(x - \xi).$$

The full solution is

$$y(x) = \frac{g}{T} \int_0^L G(x, \xi) \mu(\xi) \, d\xi,$$

representing the superposition of contributions from infinitesimal mass elements.

5.3 Green's functions for initial value problems

For initial value problems (IVPs), we specify both the function value and its derivative at a single point, rather than values at two boundary points.

Definition (Green's function for IVP). Consider $\mathcal{L}y = f(t)$ with initial conditions $y(t_0) = y'(t_0) = 0$. The *causal Green's function* satisfies $\mathcal{L}_t G(t, \tau) = \delta(t - \tau)$ with $G(t_0, \tau) = \partial_t G(t_0, \tau) = 0$.

Proposition (Structure of causal Green's function). Let y_1, y_2 be any basis of solutions to $\mathcal{L}y = 0$. The causal Green's function has the form

$$G(t, \tau) = \begin{cases} 0 & t_0 \leq t < \tau \\ C(\tau)y_1(t) + D(\tau)y_2(t) & t > \tau. \end{cases}$$

Proof. For $t < \tau$, write $G = A(\tau)y_1(t) + B(\tau)y_2(t)$. The initial conditions give

$$\begin{pmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Since y_1, y_2 are linearly independent, the matrix (the Wronskian at t_0) is non-singular, so $A = B = 0$.

For $t > \tau$, the coefficients C, D are determined by continuity at τ and the jump condition on G' . $\square$

Theorem (Causality). The solution to the IVP $\mathcal{L}y = f$ with $y(t_0) = y'(t_0) = 0$ is

$$y(t) = \int_{t_0}^t G(t, \tau) f(\tau) \, d\tau.$$

The upper limit is t (not ∞) because $G(t, \tau) = 0$ for $\tau > t$. This expresses *causality*: the solution at time t depends only on the forcing at earlier times.

Example (Simple harmonic oscillator). Consider $\ddot{y} + y = f(t)$ with $y(0) = \dot{y}(0) = 0$.

The homogeneous solutions are $\cos t$ and $\sin t$. For $t > \tau$, we write

$$G(t, \tau) = C(\tau) \cos(t - \tau) + D(\tau) \sin(t - \tau),$$

using shifted solutions for convenience. Continuity at $t = \tau$ gives $C(\tau) = 0$. The jump condition (with $\alpha = 1$) gives $D(\tau) = 1$. Thus

$$G(t, \tau) = \Theta(t - \tau) \sin(t - \tau),$$

and the solution is

$$y(t) = \int_0^t \sin(t - \tau) f(\tau) \, d\tau.$$

This is a *convolution* of $\sin t$ with the forcing $f(t)$.

6 Fourier transforms

6.1 The Fourier transform

So far, we have considered writing a function $f : S^1 \rightarrow \mathbb{C}$ (or a periodic function) as a Fourier sum

$$f(\theta) = \sum_{n \in \mathbb{Z}} \hat{f}_n e^{in\theta}, \quad \hat{f}_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\theta} f(\theta) \, d\theta.$$

What if we don't have a periodic function, but just an arbitrary function $f : \mathbb{R} \rightarrow \mathbb{C}$? Can we still do Fourier analysis? Yes!

To start with, instead of $\hat{f}_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{-in\theta} f(\theta) \, d\theta$, we define the Fourier transform as follows:

Definition (Fourier transform). The *Fourier transform* of an (absolutely integrable) function $f : \mathbb{R} \rightarrow \mathbb{C}$ is defined as

$$\tilde{f}(k) = \int_{-\infty}^{\infty} e^{-ikx} f(x) \, dx$$

for all $k \in \mathbb{R}$. We will also write $\tilde{f}(k) = \mathcal{F}[f(x)]$.

Note that for any k , we have

$$|\tilde{f}(k)| = \left| \int_{-\infty}^{\infty} e^{-ikx} f(x) \, dx \right| \leq \int_{-\infty}^{\infty} |e^{-ikx} f(x)| \, dx = \int_{-\infty}^{\infty} |f(x)| \, dx.$$

Since we have assumed that our function is absolutely integrable, this is finite, and the definition makes sense.

Proposition (Properties of the Fourier transform). Let $f, g : \mathbb{R} \rightarrow \mathbb{C}$ be absolutely integrable functions.

(i) *Linearity*: If c_1, c_2 are constants, then

$$\mathcal{F}[c_1 f(x) + c_2 g(x)] = c_1 \mathcal{F}[f(x)] + c_2 \mathcal{F}[g(x)].$$

(ii) *Translation*: For $a \in \mathbb{R}$,

$$\mathcal{F}[f(x - a)] = e^{-ika} \mathcal{F}[f(x)].$$

A translation in x -space becomes a re-phasing in k -space.

(iii) *Modulation*: For $\ell \in \mathbb{R}$,

$$\mathcal{F}[e^{i\ell x} f(x)] = \tilde{f}(k - \ell).$$

A modulation in x -space becomes a translation in k -space.

(iv) *Scaling*: For $c \neq 0$,

$$\mathcal{F}[f(cx)] = \frac{1}{|c|} \tilde{f}\left(\frac{k}{c}\right).$$

(v) *Convolution*: The *convolution* of f and g is

$$(f * g)(x) = \int_{-\infty}^{\infty} f(x-y)g(y) \, dy.$$

The Fourier transform of a convolution is the product of Fourier transforms:

$$\mathcal{F}[f * g] = \mathcal{F}[f] \cdot \mathcal{F}[g].$$

(vi) *Differentiation*: The Fourier transform turns differentiation into multiplication:

$$\mathcal{F}[f'(x)] = ik\mathcal{F}[f(x)].$$

Conversely, multiplication becomes differentiation:

$$\mathcal{F}[xf(x)] = i\tilde{f}'(k).$$

These properties allow us to break complicated Fourier transforms into simpler pieces.

Proposition (Solving ODEs via Fourier transform). Consider the differential equation $\mathcal{L}(\partial)y = f$, where

$$\mathcal{L}(\partial) = \sum_{r=0}^p c_r \frac{d^r}{dx^r}$$

is a differential operator of order p with constant coefficients. Taking Fourier transforms converts this to the algebraic equation

$$\mathcal{L}(ik)\tilde{y}(k) = \tilde{f}(k),$$

where $\mathcal{L}(ik) = \sum_{r=0}^p c_r (ik)^r$. The solution in Fourier space is

$$\tilde{y}(k) = \frac{\tilde{f}(k)}{\mathcal{L}(ik)}.$$

Example (Screened Poisson equation). Consider $\nabla^2 \phi - m^2 \phi = \rho(\mathbf{x})$ for $\phi : \mathbb{R}^n \rightarrow \mathbb{C}$. In higher dimensions, the Fourier transform is

$$\tilde{\phi}(\mathbf{k}) = \int_{\mathbb{R}^n} e^{-i\mathbf{k} \cdot \mathbf{x}} \phi(\mathbf{x}) \, d^n \mathbf{x}.$$

Using integration by parts twice, $\mathcal{F}[\nabla^2 \phi] = -|\mathbf{k}|^2 \tilde{\phi}(\mathbf{k})$. Thus the transformed equation is

$$(-|\mathbf{k}|^2 - m^2)\tilde{\phi}(\mathbf{k}) = \tilde{\rho}(\mathbf{k}),$$

giving

$$\tilde{\phi}(\mathbf{k}) = -\frac{\tilde{\rho}(\mathbf{k})}{|\mathbf{k}|^2 + m^2}.$$

Remark. Differential equations become algebraic in Fourier space. The challenge is recovering the solution in position space, which requires inverting the Fourier transform.

6.2 The Fourier inversion theorem

We need to recover a function $f(x)$ from its Fourier transform $\tilde{f}(k)$. A heuristic derivation proceeds by taking the $L \rightarrow \infty$ limit of Fourier series, treating the discrete sum as a Riemann sum that becomes an integral.

Theorem (Fourier inversion). For sufficiently well-behaved functions, the inverse Fourier transform recovers f from $\tilde{f}$:

$$f(x) = \mathcal{F}^{-1}[\tilde{f}(k)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} \tilde{f}(k) dk.$$

Remark. The heuristic derivation involves taking $L \rightarrow \infty$ and $\Delta k = 2\pi/L \rightarrow 0$ simultaneously, which is not rigorous. A proper proof requires careful analysis of convergence.

The inverse Fourier transform differs from the forward transform only by a factor of $\frac{1}{2\pi}$ and a sign change in the exponent. Some conventions distribute the 2π symmetrically by defining $\mathcal{F}[f] = \frac{1}{\sqrt{2\pi}} \int e^{-ikx} f(x) dx$. We use the asymmetric convention, which gives:

Proposition (Duality). The Fourier transform and inverse are related by

$$\mathcal{F}[f(x)](k) = 2\pi \cdot \mathcal{F}^{-1}[f(-x)](k).$$

This allows us to compute inverse Fourier transforms using forward transforms.

Remark. This duality does not occur for Fourier series, where the forward operation (integration) and inverse operation (discrete summation) are fundamentally different.

Proposition (Convolution and multiplication). The Fourier transform exchanges convolution and multiplication:

- (i) $\mathcal{F}[f * g] = \tilde{f} \cdot \tilde{g}$ (convolution becomes multiplication).
- (ii) $\mathcal{F}[f \cdot g] = \frac{1}{2\pi} \tilde{f} * \tilde{g}$ (multiplication becomes convolution).

Example (Screened Poisson: inversion). From Example 6.1, the screened Poisson equation $\nabla^2 \phi - m^2 \phi = \rho$ has Fourier solution

$$\tilde{\phi}(\mathbf{k}) = -\frac{\tilde{\rho}(\mathbf{k})}{|\mathbf{k}|^2 + m^2}.$$

Inverting gives

$$\phi(\mathbf{x}) = -\frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\mathbf{k} \cdot \mathbf{x}} \frac{\tilde{\rho}(\mathbf{k})}{|\mathbf{k}|^2 + m^2} d^n \mathbf{k}.$$

Since this is the inverse transform of a product, ϕ is a convolution of ρ with $\mathcal{F}^{-1}[(|\mathbf{k}|^2 + m^2)^{-1}]$. This is analogous to how Green's function solutions are convolutions of the Green's function with the forcing term.

6.3 Parseval's theorem for Fourier transforms

Theorem (Parseval's theorem for Fourier transforms). Let $f, g : \mathbb{R} \rightarrow \mathbb{C}$ be functions such that $\tilde{f}, \tilde{g}$ exist and $\mathcal{F}^{-1}[\tilde{f}] = f$, $\mathcal{F}^{-1}[\tilde{g}] = g$. Then

$$(f, g) = \int_{\mathbb{R}} f^*(x)g(x) dx = \frac{1}{2\pi}(\tilde{f}, \tilde{g}).$$

In particular,

$$\|f\|^2 = \frac{1}{2\pi}\|\tilde{f}\|^2.$$

Remark. The Fourier transform preserves the L^2 norm up to a factor of $\sqrt{2\pi}$. With the symmetric convention $\mathcal{F}[f] = \frac{1}{\sqrt{2\pi}} \int e^{-ikx} f(x) dx$, it would be an isometry.

Proof.

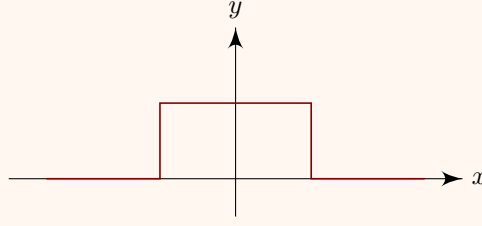
$$\begin{aligned} (f, g) &= \int_{\mathbb{R}} f^*(x)g(x) dx \\ &= \int_{-\infty}^{\infty} f^*(x) \left[\frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} \tilde{g}(k) dk \right] dx \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f^*(x) e^{ikx} dx \right] \tilde{g}(k) dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} f(x) e^{-ikx} dx \right]^* \tilde{g}(k) dk \\ &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{f}^*(k) \tilde{g}(k) dk \\ &= \frac{1}{2\pi}(\tilde{f}, \tilde{g}). \end{aligned} \quad \square$$

6.4 A word of warning

The Fourier inversion theorem requires care about convergence.

Example (Box function). Consider the box function

$$f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & |x| \geq 1. \end{cases}$$



This function is absolutely integrable, and its Fourier transform is

$$\tilde{f}(k) = \int_{-1}^1 e^{-ikx} dx = \frac{2 \sin k}{k}.$$

Proposition (Sinc is not absolutely integrable). The function $\text{sinc}(k) = \frac{\sin k}{k}$ is not absolutely integrable on $\mathbb{R}$.

Proof. On the intervals $[(n + \frac{1}{4})\pi, (n + \frac{3}{4})\pi]$, we have $|\sin k| \geq \frac{1}{\sqrt{2}}$. Thus

$$\int_{-\infty}^{\infty} \left| \frac{\sin k}{k} \right| dk \geq \sqrt{2} \sum_{n=0}^N \int_{(n+1/4)\pi}^{(n+3/4)\pi} \frac{dk}{(n+1)\pi} = \frac{\sqrt{2}}{2} \sum_{n=0}^N \frac{1}{n+1},$$

which diverges as $N \rightarrow \infty$. $\square$

Remark. This means the inverse Fourier transform $\mathcal{F}^{-1}[\tilde{f}]$ does not converge absolutely for the box function. The inversion still works, but requires more careful analysis. In practice, we often ignore such subtleties.

6.5 Fourier transformation of distributions

The Fourier transform extends to distributions, though the integrals must be interpreted carefully.

Proposition (Fourier transform of the delta function). The Fourier transform of the delta function is constant:

$$\mathcal{F}[\delta(x)] = 1.$$

Equivalently,

$$\mathcal{F}^{-1}[1] = \delta(x), \quad \text{i.e.,} \quad \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx} dk = \delta(x).$$

Remark. The integral $\int e^{ikx} dk$ diverges as an ordinary integral. It only makes sense as a distributional identity.

Proposition (Fourier transforms of translated and modulated distributions). For $a, \ell \in \mathbb{R}$:

- (i) $\mathcal{F}[\delta(x - a)] = e^{-ika}$ (translated delta becomes modulation).
- (ii) $\mathcal{F}[e^{i\ell x}] = 2\pi\delta(k - \ell)$ (plane wave becomes translated delta).
- (iii) $\mathcal{F}[\cos(\ell x)] = \pi[\delta(k - \ell) + \delta(k + \ell)]$.

Remark (Uncertainty principle). Highly localized functions in x -space (like $\delta(x)$) have spread-out Fourier transforms in k -space (like the constant 1), and vice versa. This is a manifestation of the uncertainty principle.

6.6 Linear systems and response functions

Consider a linear system (e.g., an amplifier) that transforms an input signal $I(t)$ into an output $O(t)$ by modifying the amplitudes and phases of frequency components.

Definition (Transfer function and response function). The *transfer function* $\tilde{R}(\omega)$ specifies how the system acts in frequency space:

$$\tilde{O}(\omega) = \tilde{R}(\omega)\tilde{I}(\omega).$$

The *response function* (or impulse response) is its inverse Fourier transform:

$$R(t) = \mathcal{F}^{-1}[\tilde{R}(\omega)] = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{i\omega t} \tilde{R}(\omega) d\omega.$$

Proposition (Output as convolution). The output of a linear system is the convolution of the response function with the input:

$$O(t) = (R * I)(t) = \int_{-\infty}^{\infty} R(t - u)I(u) du.$$

In particular, $R(t)$ is the response to an impulse input $I(t) = \delta(t)$.

Remark (Causality). Causality requires that the system cannot respond before receiving input, so $R(t) = 0$ for $t < 0$. For input starting at $t = 0$:

$$O(t) = \int_0^t R(t - u)I(u) du.$$

This is the same form as solutions via Green's functions for IVPs, with $R(t)$ playing the role of the causal Green's function.

6.7 General form of transfer function

Suppose the system is governed by an ODE with the input as forcing:

$$\mathcal{L}_m O(t) = I(t), \quad \text{where} \quad \mathcal{L}_m = \sum_{i=0}^m a_i \frac{d^i}{dt^i}$$

with constant coefficients $a_i \in \mathbb{C}$.

Proposition (Transfer function for ODE systems). For the system $\mathcal{L}_m O = I$, the transfer function is

$$\tilde{R}(\omega) = \frac{1}{\sum_{j=0}^m a_j (i\omega)^j}.$$

If the denominator has roots $c_j \in \mathbb{C}$ with multiplicities k_j , partial fractions give

$$\tilde{R}(\omega) = \sum_{j=1}^J \sum_{r=1}^{k_j} \frac{\Gamma_{rj}}{(i\omega - c_j)^r}$$

for constants $\Gamma_{rj} \in \mathbb{C}$.

To find the response function, we need inverse Fourier transforms of terms $(i\omega - \alpha)^{-p}$.

Lemma (Inverse Fourier transform of rational functions). For $\text{Re}(\alpha) < 0$ and $p \geq 0$, define

$$h_p(t) = \begin{cases} \frac{t^p}{p!} e^{\alpha t} & t > 0 \\ 0 & t \leq 0. \end{cases}$$

Then

$$\mathcal{F}[h_p(t)] = \frac{1}{(i\omega - \alpha)^{p+1}}.$$

Proof. For $p = 0$: $\mathcal{F}[h_0] = \int_0^\infty e^{(\alpha - i\omega)t} dt = \frac{1}{i\omega - \alpha}$, valid when $\text{Re}(\alpha) < 0$.

For $p \geq 1$: use $\mathcal{F}[t \cdot g(t)] = i \frac{d}{d\omega} \mathcal{F}[g]$ inductively. $\square$

Remark (Stability). The response function is a linear combination of terms $h_p(t)$, which vanish for $t < 0$ (causality). If any root c_j has $\text{Re}(c_j) \geq 0$, the system is unstable and better analysed via Laplace transforms.

Example (ODE on the real line). Consider $y'' - A^2 y = -f(x)$ on $\mathbb{R}$ with $y, y' \rightarrow 0$ as $|x| \rightarrow \infty$. Taking Fourier transforms:

$$\tilde{y}(k) = \frac{\tilde{f}(k)}{k^2 + A^2}.$$

Since $\mathcal{F}\left[\frac{1}{2\mu} e^{-\mu|x|}\right] = \frac{1}{\mu^2 + k^2}$ for $\mu > 0$, the solution is

$$y(x) = \frac{1}{2A} \int_{-\infty}^{\infty} e^{-A|x-u|} f(u) du.$$

Remark. Fourier analysis works over various abelian groups: S^1 (Fourier series), $\mathbb{R}$ (Fourier transform), and $\mathbb{Z}_N$ (discrete Fourier transform, discussed next).

6.8 The discrete Fourier transform

In practice, we often only have sampled data $\{f(x_j)\}_{j=0}^{N-1}$ rather than a closed-form function. Approximating the Fourier integral by a Riemann sum leads to the discrete Fourier transform.

Definition (Discrete Fourier transform). Let $\omega = e^{2\pi i/N}$ be a primitive N th root of unity. For data $(g_0, g_1, \dots, g_{N-1}) \in \mathbb{C}^N$, the *discrete Fourier transform (DFT)* is

$$F(m) = \frac{1}{N} \sum_{j=0}^{N-1} g_j \omega^{-jm}, \quad m = 0, 1, \dots, N-1.$$

Remark. Since $\omega^N = 1$, we have $F(m+N) = F(m)$, so the DFT is periodic. This discretization loses information about structures varying faster than the sampling interval.

To understand the DFT algebraically, consider functions on the group $G = \{1, \omega, \omega^2, \dots, \omega^{N-1}\} \cong \mathbb{Z}_N$. The space of functions $g : G \rightarrow \mathbb{C}$ is isomorphic to $\mathbb{C}^N$ with inner product

$$(f, g) = \frac{1}{N} \sum_{j=0}^{N-1} f^*(\omega^j) g(\omega^j).$$

Proposition (Orthonormal basis for DFT). Define $e_m : G \rightarrow \mathbb{C}$ by $e_m(\omega^j) = \omega^{jm}$. Then $\{e_0, e_1, \dots, e_{N-1}\}$ is an orthonormal basis:

$$(e_n, e_m) = \frac{1}{N} \sum_{j=0}^{N-1} \omega^{j(m-n)} = \delta_{nm}.$$

Proof. For $n = m$: $(e_m, e_m) = \frac{1}{N} \sum_{j=0}^{N-1} 1 = 1$.

For $n \neq m$: the sum is geometric with ratio $\omega^{m-n} \neq 1$, giving $\frac{1}{N} \cdot \frac{1-\omega^{(m-n)N}}{1-\omega^{m-n}} = 0$ since $\omega^N = 1$. $\square$

Theorem (DFT inversion). The DFT coefficients $F(m) = (e_m, g)$ allow exact reconstruction:

$$g_j = \sum_{m=0}^{N-1} F(m) \omega^{jm}.$$

Remark. The DFT is exactly analogous to the continuous Fourier transform, but everything is discrete and finite sums replace integrals, eliminating convergence issues.

6.9 The fast Fourier transform*

Computing the DFT directly requires $O(N^2)$ operations. The *fast Fourier transform (FFT)* reduces this to $O(N \log N)$.

Remark (Historical motivation). The FFT was developed during the Cold War to analyze seismic data for detecting underground nuclear tests. With millions of data points, $O(N^2)$ was computationally infeasible.

Theorem (Fast Fourier transform). When $N = 2^n$, the DFT can be computed in $O(N \log N)$ operations using the recursion:

$$F(m) = \frac{1}{2} \left[\tilde{G}(m) + \omega^{-m} \tilde{H}(m) \right],$$

where $\tilde{G}$ and $\tilde{H}$ are length- $N/2$ DFTs of the even- and odd-indexed inputs respectively.

Proof sketch. Split the sum into even and odd terms: if $G(\eta^k) = g(\omega^{2k})$ and $H(\eta^k) = g(\omega^{2k+1})$ where $\eta = \omega^2$ is an $(N/2)$ th root of unity, then

$$F(m) = \frac{1}{2} \left[\tilde{G}(m) + \omega^{-m} \tilde{H}(m) \right].$$

If P_N denotes the operation count, then $P_N = 2P_{N/2} + O(N)$, which solves to $P_N = O(N \log N)$. $\square$

Remark. The FFT is purely a computational optimization with no new mathematics. It is ubiquitous in signal processing, image compression, and scientific computing.

7 More partial differential equations

7.1 Well-posedness

Different PDEs require different types of boundary/initial conditions:

- *Laplace's equation* $\nabla^2\phi = 0$: Dirichlet ($\phi|_{\partial\Omega} = f$) or Neumann ($\mathbf{n} \cdot \nabla\phi|_{\partial\Omega} = g$).
- *Heat equation* $\partial_t\phi = \kappa\nabla^2\phi$: boundary conditions plus initial condition $\phi|_{t=0}$.
- *Wave equation* $\partial_t^2\phi = c^2\nabla^2\phi$: boundary conditions plus initial data for both $\phi|_{t=0}$ and $\partial_t\phi|_{t=0}$.

Definition (Cauchy data). The *Cauchy data* for a PDE consists of values of ϕ (and possibly its derivatives) specified on a codimension-1 surface.

Definition (Well-posed problem). A PDE problem is *well-posed* (in the sense of Hadamard) if:

- (i) A solution exists;
- (ii) The solution is unique;
- (iii) The solution depends continuously on the Cauchy data.

Example (Heat equation: well-posed). The forward heat equation $\partial_t\phi = \kappa\nabla^2\phi$ is well-posed: solutions smooth out over time, so small changes in initial data produce small changes in the solution. However, the *backward* heat equation is ill-posed: small perturbations grow exponentially as we evolve backward in time.

Example (Laplace with Cauchy data: ill-posed). Consider $\nabla^2\phi = 0$ on the upper half-plane with $\phi(x, 0) = 0$ and $\partial_y\phi(x, 0) = g(x)$. For $g(x) = \frac{\sin(Ax)}{A}$, the solution is $\phi = \frac{\sin(Ax)\sinh(Ay)}{A^2}$. As $A \rightarrow \infty$: $g(x) \rightarrow 0$ pointwise, but $\phi(\frac{\pi}{2A}, y) = \frac{\sinh(Ay)}{A^2}$ becomes unbounded. This violates continuous dependence, so the problem is ill-posed.

7.2 Method of characteristics

Curves in the plane

Suppose we have a curve on the plane $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{R}^2$ given by $s \mapsto (x(s), y(s))$.

Definition (Tangent vector). The tangent vector to a smooth curve C given by $\mathbf{x} : \mathbb{R} \rightarrow \mathbb{R}^2$ with $\mathbf{x}(s) = (x(s), y(s))$ is

$$\left(\frac{dx}{ds}, \frac{dy}{ds} \right).$$

Remark. If $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ is a scalar function, then its value along a curve C parametrized by $\mathbf{x}(s) = (x(s), y(s))$ is simply the composition $\phi(x(s), y(s))$.

A vector field assigns a vector to each point in the plane. Such a vector field naturally defines a family of curves: intuitively, if we imagine the vector field as describing the flow of water in a river, we can obtain a curve by starting at a point and flowing along with the water.

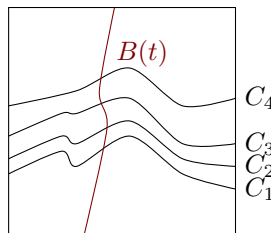
Definition (Integral curve). Let $\mathbf{V} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a vector field with $\mathbf{V}(x, y) = (V_1(x, y), V_2(x, y))$. An *integral curve* of $\mathbf{V}$ is a curve $\mathbf{x}(s) = (x(s), y(s))$ whose tangent vector at each point equals the value of $\mathbf{V}$ at that point, i.e.,

$$\frac{dx}{ds} = V_1(x, y), \quad \frac{dy}{ds} = V_2(x, y).$$

For sufficiently regular vector fields, the integral curves fill the plane. To parametrize which curve we are on, we introduce a second parameter t .

Definition (Transverse curve). A curve $B(t) = (x(t), y(t))$ is *transverse* to a family of integral curves if B is nowhere parallel to the integral curves, i.e., the tangent vector of B is never parallel to the vector field $\mathbf{V}$.

Given a transverse curve B , we can label each integral curve by the value of t at which it intersects B . This is illustrated in the following figure:



We can thus describe the family of integral curves by $(x(s, t), y(s, t))$, where s is the parameter along each curve and t labels which curve we are on.

Proposition (Characteristic coordinates). Let $(x(s, t), y(s, t))$ parametrize

a family of integral curves as above. If the Jacobian

$$J = \frac{\partial x}{\partial s} \frac{\partial y}{\partial t} - \frac{\partial x}{\partial t} \frac{\partial y}{\partial s}$$

is non-zero, then we can invert this to find (s, t) as a function of (x, y) . In other words, at any point (x, y) , we can determine which curve we are on and how far along the curve we are. The pair (s, t) forms a new coordinate system for points in $\mathbb{R}^2$, called *characteristic coordinates*.

By choosing an appropriate vector field $\mathbf{V}$, we can make differential equations much easier to solve in characteristic coordinates.

The method of characteristics

Consider a first-order linear partial differential equation of the form

$$a(x, y) \frac{\partial \phi}{\partial x} + b(x, y) \frac{\partial \phi}{\partial y} = 0,$$

where $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ is the unknown function and $a, b : \mathbb{R}^2 \rightarrow \mathbb{R}$ are given coefficient functions.

Definition (Characteristic vector field and equations). The *characteristic vector field* associated with the PDE above is

$$\mathbf{V}(x, y) = \begin{pmatrix} a(x, y) \\ b(x, y) \end{pmatrix}.$$

The *characteristic equations* are the ODEs that determine the integral curves of $\mathbf{V}$:

$$\frac{dx}{ds} = a(x, y), \quad \frac{dy}{ds} = b(x, y).$$

These integral curves are called the *characteristics* of the PDE.

Using the characteristic vector field, the PDE can be written compactly as

$$\mathbf{V} \cdot \nabla \phi = 0.$$

Theorem (Method of characteristics). Let ϕ satisfy the PDE $\mathbf{V} \cdot \nabla \phi = 0$. Then ϕ is constant along each characteristic curve. More precisely, if $(x(s), y(s))$ is a characteristic curve, then

$$\frac{d}{ds} \phi(x(s), y(s)) = 0.$$

Proof. By the chain rule, along a characteristic curve $(x(s), y(s))$ we have

$$\frac{d\phi}{ds} = \frac{dx}{ds} \frac{\partial \phi}{\partial x} + \frac{dy}{ds} \frac{\partial \phi}{\partial y} = a(x, y) \frac{\partial \phi}{\partial x} + b(x, y) \frac{\partial \phi}{\partial y} = \mathbf{V} \cdot \nabla \phi = 0.$$

Thus ϕ does not change along characteristics. $\square$

Remark (Solution procedure). To solve the PDE with boundary data specified along a transverse curve B , we proceed as follows:

- (i) Parametrize B by t , so that B corresponds to $s = 0$ in characteristic coordinates.
- (ii) Suppose the boundary condition is $\phi(0, t) = h(t)$ for some given function h .
- (iii) Since ϕ is constant along characteristics, the solution in characteristic coordinates is simply $\phi(s, t) = h(t)$.
- (iv) To obtain $\phi(x, y)$, invert the characteristic equations to find t as a function of (x, y) .

We illustrate this method with several examples.

Example (A simple first-order PDE). Consider the PDE

$$\frac{\partial \phi}{\partial x} = 0$$

with boundary condition $\phi(0, y) = f(y)$.

Direct solution. The PDE says that ϕ does not depend on x , so immediately we have $\phi(x, y) = f(y)$.

Solution via characteristics. We apply the method of characteristics for practice. The characteristic vector field is

$$\mathbf{V} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

The characteristic equations are

$$\frac{dx}{ds} = 1, \quad \frac{dy}{ds} = 0.$$

Integrating, we obtain

$$x = s + c, \quad y = d,$$

where c and d are constants of integration. We choose the boundary curve B to be $x = 0$, parametrized by $y = t$. Setting $s = 0$ along B , we find the characteristics are

$$x = s, \quad y = t.$$

The boundary condition $\phi(0, y) = f(y)$ becomes $\phi(0, t) = f(t)$ in characteristic coordinates. Since ϕ is constant along characteristics, the solution is

$$\phi(s, t) = f(t).$$

Converting back to (x, y) coordinates using $t = y$, we obtain

$$\phi(x, y) = f(y),$$

in agreement with the direct solution.

Example (First-order PDE with exponential coefficient). Consider the PDE

$$e^x \frac{\partial \phi}{\partial x} + \frac{\partial \phi}{\partial y} = 0$$

with boundary condition $\phi(x, 0) = \cosh x$.

The characteristic vector field is

$$\mathbf{V} = \begin{pmatrix} e^x \\ 1 \end{pmatrix}.$$

The characteristic equations are

$$\frac{dx}{ds} = e^x, \quad \frac{dy}{ds} = 1.$$

Integrating the first equation by separation of variables and the second directly, we obtain

$$e^{-x} = -s + c, \quad y = s + d,$$

where c and d are constants of integration.

We choose the boundary curve B to be $y = 0$, parametrized by $x = t$. Setting $s = 0$ along B gives $c = e^{-t}$ and $d = 0$, so the characteristics are

$$e^{-x} = -s + e^{-t}, \quad y = s.$$

The boundary condition $\phi(x, 0) = \cosh x$ becomes $\phi(0, t) = \cosh t$ in characteristic coordinates. Since ϕ is constant along characteristics,

$$\phi(s, t) = \cosh t.$$

To express the solution in (x, y) coordinates, we solve for t . From $y = s$ and $e^{-x} = -s + e^{-t}$, we get $e^{-t} = y + e^{-x}$, hence $t = -\ln(y + e^{-x})$. Thus

$$\phi(x, y) = \cosh[-\ln(y + e^{-x})].$$

Example (Inhomogeneous first-order PDE). Consider the inhomogeneous PDE

$$\frac{\partial \phi}{\partial x} + 2 \frac{\partial \phi}{\partial y} = ye^x$$

with boundary condition $\phi(x, 0) = \sin x$.

The method of characteristics extends to inhomogeneous equations. The characteristic vector field is

$$\mathbf{V} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

The characteristic equations are

$$\frac{dx}{ds} = 1, \quad \frac{dy}{ds} = 2.$$

Integrating, we obtain

$$x = s + c, \quad y = 2s + d.$$

We choose the boundary curve B to be the line $y = x$, parametrized by $x = t$ (equivalently, $y = t$). Setting $s = 0$ along B gives $c = t$ and $d = t$, so

$$x = s + t, \quad y = 2s + t.$$

Inverting, we find

$$s = y - x, \quad t = 2x - y.$$

Along a characteristic, the PDE becomes an ODE in s :

$$\frac{d\phi}{ds} = \mathbf{V} \cdot \nabla \phi = ye^x = (2s + t)e^{s+t}.$$

The boundary condition $\phi(x, x) = \sin x$ becomes $\phi(0, t) = \sin t$. Integrating the ODE with this initial condition gives

$$\phi(s, t) = (2 - t)e^t(1 - e^s) + \sin t + 2se^{s+t}.$$

Converting to (x, y) coordinates using $s = y - x$ and $t = 2x - y$, we obtain

$$\phi(x, y) = (2 - 2x + y)e^{2x-y} + \sin(2x - y) + (y - 2)e^x.$$

Remark (Well-posedness and the choice of boundary curve). The boundary (or Cauchy) data must be specified on a curve B that intersects each characteristic curve exactly once. Several important observations follow:

- (i) If B intersects the same characteristic multiple times, then the boundary data cannot be specified freely along B —the values at the points of intersection must be compatible with the PDE. For a homogeneous equation, the solution is constant along each characteristic, so if the boundary data prescribes different values at two points on the same characteristic, no solution exists.
- (ii) If B does not intersect all characteristics, then even if the data is compatible, the solution will not be unique: the value of ϕ is undetermined along characteristics that do not meet B .
- (iii) If B intersects each characteristic curve transversely (i.e., B is a transverse curve in the sense defined earlier), then the problem is well-posed and there exists a unique solution, at least in a neighbourhood of B .

An important feature of the method is that information is transported along characteristics but never between them.

7.3 Characteristics for 2nd order partial differential equations

Whether the method of characteristics works for a 2nd order partial differential equation, or even higher order ones, depends on the “type” of the differential equation.

To classify these differential equations, we need to define

Definition (Symbol and principal part). Let $\mathcal{L}$ be the general 2nd order differential operator on $\mathbb{R}^n$. We can write it as

$$\mathcal{L} = \sum_{i,j=1}^n a^{ij}(x) \frac{\partial^2}{\partial x^i \partial x^j} + \sum_{i=1}^n b^i(x) \frac{\partial}{\partial x^i} + c(x),$$

where $a^{ij}(x), b^i(x), c(x) \in \mathbb{R}$ and $a^{ij} = a^{ji}$ (wlog).

We define the *symbol* $\sigma(\mathbf{k}, x)$ of $\mathcal{L}$ to be

$$\sigma(\mathbf{k}, x) = \sum_{i,j=1}^n a^{ij}(x) k_i k_j + \sum_{i=1}^n b^i(x) k_i + c(x).$$

So we just replace the derivatives by the variable k .

The *principal part* of the symbol is the leading term

$$\sigma^p(\mathbf{k}, x) = \sum_{i,j=1}^n a^{ij}(x) k_i k_j.$$

Example (Symbol of Laplacian and heat operator). If $\mathcal{L} = \nabla^2$, then

$$\sigma(\mathbf{k}, x) = \sigma^p(\mathbf{k}, x) = \sum_{i=1}^n (k_i)^2.$$

If $\mathcal{L}$ is the heat operator (where we think of the last coordinate x^n as the time, and others as space), then the operator is given by

$$\mathcal{L} = \frac{\partial}{\partial x^n} - \sum_{i=1}^{n-1} \frac{\partial^2}{\partial x^{i2}}.$$

The symbol is then

$$\sigma(\mathbf{k}, x) = k_n - \sum_{i=1}^{n-1} (k_i)^2,$$

and the principal part is

$$\sigma^p(\mathbf{k}, x) = - \sum_{i=1}^{n-1} (k_i)^2.$$

Remark. The symbol is closely related to the Fourier transform of the differential operator, since both turn differentiation into multiplication. Indeed, they are equal when the coefficients are constant. However, the symbol is defined for arbitrary differential operators, including those with non-constant coefficients.

For each x , the principal part of the symbol can be written as the quadratic form

$$\sigma^p(\mathbf{k}, x) = \mathbf{k}^T A(x) \mathbf{k},$$

where $A(x)$ is the symmetric matrix with entries $a^{ij}(x)$. Since $A(x)$ is real and symmetric, it has real eigenvalues, which allows us to classify the differential operator based on the signs of these eigenvalues.

Definition (Elliptic, hyperbolic, ultra-hyperbolic and parabolic differential operators). Let $\mathcal{L}$ be a differential operator. We say $\mathcal{L}$ is

- *elliptic at x* if all eigenvalues of $A(x)$ have the same sign. Equivalently, if $\sigma^p(\cdot, x)$ is a definite quadratic form;
- *hyperbolic at x* if all but one eigenvalues of $A(x)$ have the same sign;
- *ultra-hyperbolic at x* if $A(x)$ has more than one eigenvalues of each sign;
- *parabolic at x* if $A(x)$ has a zero eigenvalue, i.e. $\sigma^p(\cdot, x)$ is degenerate.

We say $\mathcal{L}$ is elliptic if $\mathcal{L}$ is elliptic at all x , and similarly for the other terms.

Remark. The terminology is motivated by conic sections: the equation $\mathbf{k}^T A \mathbf{k} = 1$ defines an ellipse, hyperbola, or parabola depending on the signs of the eigenvalues of A .

Example (PDE classification in two dimensions). Let

$$\mathcal{L} = a(x, y) \frac{\partial^2}{\partial x^2} + 2b(x, y) \frac{\partial^2}{\partial x \partial y} + c(x, y) \frac{\partial^2}{\partial y^2} + d(x, y) \frac{\partial}{\partial x} + e(x, y) \frac{\partial}{\partial y} + f(x, y).$$

Then the principal part of the symbol is

$$\sigma^p(\mathbf{k}, x) = \begin{pmatrix} k_x & k_y \end{pmatrix} \begin{pmatrix} a(x, y) & b(x, y) \\ b(x, y) & c(x, y) \end{pmatrix} \begin{pmatrix} k_x \\ k_y \end{pmatrix}$$

Then $\mathcal{L}$ is elliptic at x if $b^2 - ac < 0$; hyperbolic if $b^2 - ac > 0$; and parabolic if $b^2 - ac = 0$.

Note that since we only have two dimensions and hence only two eigenvalues, we cannot possibly have an ultra-hyperbolic equation.

Characteristic surfaces

Definition (Characteristic surface). Given a second-order differential operator $\mathcal{L}$ with principal symbol matrix $A(x)$, let

$$f(x^1, x^2, \dots, x^n) = 0$$

define a surface $C \subseteq \mathbb{R}^n$. We say C is a *characteristic surface* if

$$\sigma^p(\nabla f, x) = (\nabla f)^T A(x) (\nabla f) = \sum_{i,j=1}^n a^{ij}(x) \frac{\partial f}{\partial x^i} \frac{\partial f}{\partial x^j} = 0.$$

In two dimensions, a characteristic surface is simply a curve, called a *characteristic curve*.

Remark. The characteristic equation $(\nabla f)^T A(\nabla f) = 0$ constrains the possible values of ∇f . Since ∇f is normal to the surface C , this determines the allowed normal directions at each point. Characteristic surfaces are constructed by integrating these normal directions.

We now analyze the existence of characteristics for each type of operator.

Proposition (Characteristics of elliptic operators). An elliptic operator has no real characteristic surfaces.

Proof. For an elliptic operator, all eigenvalues of A have the same sign, so $(\nabla f)^T A(\nabla f)$ is a definite quadratic form. The only solution to $(\nabla f)^T A(\nabla f) = 0$ is $\nabla f = 0$, which does not define a surface. $\square$

Remark. Since elliptic operators have no real characteristics, the method of characteristics is not useful for equations like Laplace's equation, at least when working with real numbers.

Proposition (Characteristics of parabolic operators). A parabolic operator (with exactly one zero eigenvalue) has a unique family of characteristic surfaces.

Proof. Assume A has exactly one zero eigenvector $\mathbf{n}$ (normalized so that $\mathbf{n} \cdot \mathbf{n} = 1$), with all other eigenvalues having the same sign. Then $A\mathbf{n} = \mathbf{0}$.

Decompose ∇f into components parallel and perpendicular to $\mathbf{n}$:

$$\nabla f = \mathbf{n}(\mathbf{n} \cdot \nabla f) + \nabla_{\perp} f,$$

where $\nabla_{\perp} f = \nabla f - \mathbf{n}(\mathbf{n} \cdot \nabla f)$ is the component orthogonal to $\mathbf{n}$.

Computing the characteristic condition:

$$\begin{aligned} (\nabla f)^T A(\nabla f) &= [\mathbf{n}(\mathbf{n} \cdot \nabla f) + \nabla_{\perp} f]^T A[\mathbf{n}(\mathbf{n} \cdot \nabla f) + \nabla_{\perp} f] \\ &= (\nabla_{\perp} f)^T A(\nabla_{\perp} f), \end{aligned}$$

since terms involving $A\mathbf{n}$ vanish. The quadratic form $(\nabla_\perp f)^T A(\nabla_\perp f)$ is definite on the subspace orthogonal to $\mathbf{n}$, so the only solution is $\nabla_\perp f = 0$. Thus ∇f must be parallel to $\mathbf{n}$, giving a unique characteristic direction at each point. $\square$

Proposition (Characteristics of hyperbolic operators). A hyperbolic operator has two families of characteristic surfaces through each point.

Proof. Assume all but one eigenvalues of A are positive, and let $-\lambda$ (with $\lambda > 0$) be the unique negative eigenvalue with corresponding unit eigenvector $\mathbf{n}$.

Using the decomposition $\nabla f = \mathbf{n}(\mathbf{n} \cdot \nabla f) + \nabla_\perp f$, the characteristic condition becomes:

$$\begin{aligned} 0 &= (\nabla f)^T A(\nabla f) \\ &= -\lambda(\mathbf{n} \cdot \nabla f)^2 + (\nabla_\perp f)^T A(\nabla_\perp f). \end{aligned}$$

Solving for the component along $\mathbf{n}$:

$$\mathbf{n} \cdot \nabla f = \pm \sqrt{\frac{(\nabla_\perp f)^T A(\nabla_\perp f)}{\lambda}}.$$

The two choices of sign give two distinct families of characteristic surfaces through each point. $\square$

In two dimensions, we can find the characteristic curves explicitly.

Proposition (Characteristic curves in two dimensions). Consider a second-order operator in two dimensions with principal part

$$a(x, y) \frac{\partial^2}{\partial x^2} + 2b(x, y) \frac{\partial^2}{\partial x \partial y} + c(x, y) \frac{\partial^2}{\partial y^2}.$$

The characteristic curves $y = y(x)$ satisfy

$$\frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - ac}}{a}.$$

Proof. Let $f(x, y) = 0$ define a characteristic curve, which we write as $y = y(x)$. Since f is constant along the curve, the chain rule gives

$$0 = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \frac{dy}{dx}.$$

The characteristic condition $a f_x^2 + 2b f_x f_y + c f_y^2 = 0$ can be solved for the ratio f_x/f_y , which yields the stated formula for dy/dx . $\square$

Remark. This formula makes explicit how the type of the PDE determines the number of characteristics:

- If $b^2 - ac > 0$ (hyperbolic), there are two distinct real characteristic curves through each point.

- If $b^2 - ac = 0$ (parabolic), there is exactly one characteristic curve through each point.
- If $b^2 - ac < 0$ (elliptic), there are no real characteristics.

Example (Variable type PDE). Consider

$$\partial_y^2 \phi - xy \partial_x^2 \phi = 0$$

on $\mathbb{R}^2$. Then $a = -xy, b = 0, c = 1$. So $b^2 - ac = xy$. So the type is elliptic if $xy < 0$, hyperbolic if $xy > 0$, and parabolic if $xy = 0$. In the regions where it is hyperbolic, we find

$$\frac{-b \pm \sqrt{b^2 - ac}}{a} = \pm \frac{1}{\sqrt{xy}}.$$

Hence the two characteristics are given by

$$\frac{dy}{dx} = \pm \frac{1}{\sqrt{xy}}.$$

This has a solution

$$\frac{1}{3}y^{3/2} \pm x^{1/2} = c.$$

We now let

$$\begin{aligned} u &= \frac{1}{3}y^{3/2} + x^{1/2}, \\ v &= \frac{1}{3}y^{3/2} - x^{1/2}. \end{aligned}$$

Then the equation becomes

$$\frac{\partial^2 \phi}{\partial u \partial v} + \text{lower order terms} = 0.$$

Example (Wave equation, d'Alembert). Consider the wave equation

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \frac{\partial^2 \phi}{\partial x^2} = 0$$

on $\mathbb{R}^{1,1}$. Then the equation is hyperbolic everywhere, and the characteristic curves are $x \pm ct = \text{const}$. Let's look for a solution to the wave equation that obeys

$$\phi(x, 0) = f(x), \quad \partial_t \phi(x, 0) = g(x).$$

Now put $u = x - ct, v = x + ct$. Then the wave equation becomes

$$\frac{\partial^2}{\partial u \partial v} = 0.$$

So the general solution to this is

$$\phi(x, t) = G(u) + H(v) = G(x - ct) + H(x + ct).$$

The initial conditions now fix these functions

$$f(x) = G(x) + H(x), \quad g(x) = -cG'(x) + cH'(x).$$

Solving these, we find

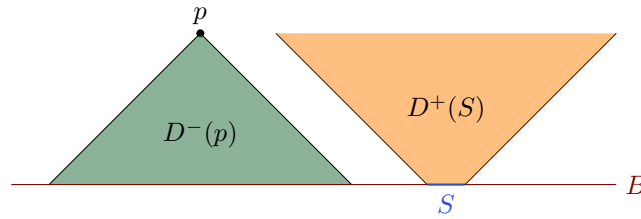
$$\phi(x, t) = \frac{1}{2}[f(x - ct) + f(x + ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} g(y) \, dy.$$

This is d'Alembert's solution to the 1 + 1 dimensional wave equation.

The d'Alembert solution reveals important concepts about how information propagates in hyperbolic equations.

Definition (Domain of dependence). The *(past) domain of dependence* of the solution at a point (x, t) , denoted $D^-(x, t)$, is the set of points on the initial surface whose data determines the value of the solution at (x, t) . For the wave equation, this is the interval $[x - ct, x + ct]$ on the initial line $t = 0$.

Definition (Range of influence). The *range of influence* of initial data at a point x_0 , denoted $D^+(x_0)$, is the set of points in spacetime where the solution can be affected by the data at x_0 . For the wave equation, this is the region $\{(x, t) : x_0 - ct \leq x \leq x_0 + ct, t \geq 0\}$.



Remark. The domain of dependence and range of influence are bounded by the characteristic curves. This shows that disturbances in the wave equation propagate with finite speed c .

7.4 Green's functions for PDEs on $\mathbb{R}^n$

Green's functions for the heat equation

Consider the heat equation on $\mathbb{R}^n$:

$$\partial_t \phi = D \nabla^2 \phi,$$

where $\phi : \mathbb{R}^n \times [0, \infty) \rightarrow \mathbb{R}$ and $D > 0$ is the diffusion constant. We seek solutions subject to the initial condition

$$\phi(\mathbf{x}, 0) = f(\mathbf{x}).$$

To solve this, we take the Fourier transform with respect to the spatial variables. For simplicity, we first consider the case $n = 1$. Writing $\mathcal{F}[\phi(x, t)] = \tilde{\phi}(k, t)$, the heat equation becomes

$$\partial_t \tilde{\phi}(k, t) = -Dk^2 \tilde{\phi}(k, t),$$

with the initial condition $\tilde{\phi}(k, 0) = \tilde{f}(k)$. For each fixed k , this is an ordinary differential equation in t with solution

$$\tilde{\phi}(k, t) = \tilde{f}(k) e^{-Dk^2 t}.$$

Taking the inverse Fourier transform:

$$\phi(x, t) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{ikx} \tilde{f}(k) e^{-Dk^2 t} dk = f * \mathcal{F}^{-1}[e^{-Dk^2 t}].$$

Using the Fourier transform of a Gaussian,

$$\mathcal{F}[e^{-a^2 x^2}] = \frac{\sqrt{\pi}}{a} e^{-k^2/4a^2},$$

with $a^2 = 1/(4Dt)$, we find

$$\mathcal{F}^{-1}[e^{-Dk^2 t}] = \frac{1}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right).$$

Definition (Fundamental solution of the heat equation). The *fundamental solution* (or *heat kernel*) of the heat equation in n spatial dimensions is

$$S_n(\mathbf{x}, t) = \frac{1}{(4\pi Dt)^{n/2}} \exp\left(-\frac{|\mathbf{x}|^2}{4Dt}\right).$$

In one dimension, we write $S_1(x, t) = (4\pi Dt)^{-1/2} \exp(-x^2/4Dt)$.

Proposition (Solution of the heat equation via convolution). The solution to the heat equation with initial condition $\phi(\mathbf{x}, 0) = f(\mathbf{x})$ is

$$\phi(\mathbf{x}, t) = \int_{\mathbb{R}^n} f(\mathbf{y}) S_n(\mathbf{x} - \mathbf{y}, t) d^n y.$$

Example (Heat equation with delta initial data). Consider the initial condition $f(x) = \phi_0 \delta(x)$, representing an initial temperature distribution

concentrated at the origin. The solution is

$$\phi(x, t) = \frac{\phi_0}{\sqrt{4\pi Dt}} \exp\left(-\frac{x^2}{4Dt}\right).$$

As time evolves, the initial delta function spreads into a Gaussian that becomes shorter and broader.

Remark (Infinite propagation speed). For the heat equation, if the initial data is supported at a single point (e.g., a delta function at the origin), then for any $t > 0$, no matter how small, the solution $\phi(x, t)$ is non-zero everywhere. This means that information propagates instantaneously to all of space. This behaviour contrasts sharply with the wave equation, where disturbances propagate at finite speed c . The infinite propagation speed is a consequence of the heat equation being parabolic.

We now consider the *inhomogeneous* (or *forced*) heat equation

$$\partial_t \phi - D\nabla^2 \phi = F(\mathbf{x}, t),$$

with homogeneous initial conditions $\phi|_{t=0} = 0$. Physically, F represents an external heat source.

Remark (Superposition principle). By linearity, if we can solve both the homogeneous equation with inhomogeneous initial data and the inhomogeneous equation with homogeneous initial data, then the general solution is their sum. Specifically, if ϕ_H solves $\partial_t \phi_H = D\nabla^2 \phi_H$ with $\phi_H|_{t=0} = f$, and ϕ_F solves $\partial_t \phi_F - D\nabla^2 \phi_F = F$ with $\phi_F|_{t=0} = 0$, then $\phi = \phi_H + \phi_F$ solves the full problem.

We solve the forced equation by Fourier transform. In one spatial dimension, the transformed equation is

$$\partial_t \tilde{\phi}(k, t) + Dk^2 \tilde{\phi}(k, t) = \tilde{F}(k, t),$$

with initial condition $\tilde{\phi}(k, 0) = 0$. This is a first-order ODE in t . Using an integrating factor:

$$\frac{\partial}{\partial t} [e^{Dk^2 t} \tilde{\phi}(k, t)] = e^{Dk^2 t} \tilde{F}(k, t),$$

which integrates to

$$\tilde{\phi}(k, t) = e^{-Dk^2 t} \int_0^t e^{Dk^2 u} \tilde{F}(k, u) \, du.$$

Definition (Green's function for the heat equation). The *Green's function* $G_n(\mathbf{x}, t; \mathbf{y}, \tau)$ for the heat equation in n spatial dimensions is the solution to

$$[\partial_t - D\nabla_{\mathbf{x}}^2] G_n(\mathbf{x}, t; \mathbf{y}, \tau) = \delta^{(n)}(\mathbf{x} - \mathbf{y}) \delta(t - \tau),$$

with the initial condition $G_n(\mathbf{x}, 0; \mathbf{y}, \tau) = 0$ for $\tau > 0$.

We compute G_1 in one spatial dimension by Fourier transform. The Fourier transform with respect to x gives

$$\tilde{G}(k, t; y, \tau) = \Theta(t - \tau) e^{-iky} e^{-Dk^2(t-\tau)}.$$

Taking the inverse Fourier transform (which is the inverse transform of a Gaussian with a phase shift):

$$G_1(x, t; y, \tau) = \frac{\Theta(t - \tau)}{\sqrt{4\pi D(t - \tau)}} \exp\left(-\frac{(x - y)^2}{4D(t - \tau)}\right) = \Theta(t - \tau) S_1(x - y, t - \tau).$$

Proposition (Green's function for the heat equation). The Green's function for the heat equation in n dimensions is

$$G_n(\mathbf{x}, t; \mathbf{y}, \tau) = \Theta(t - \tau) S_n(\mathbf{x} - \mathbf{y}, t - \tau),$$

where S_n is the fundamental solution. The solution to the forced heat equation

$$\partial_t \phi - D \nabla^2 \phi = F(\mathbf{x}, t), \quad \phi(\mathbf{x}, 0) = 0$$

is given by

$$\phi(\mathbf{x}, t) = \int_0^t \int_{\mathbb{R}^n} F(\mathbf{y}, \tau) G_n(\mathbf{x}, t; \mathbf{y}, \tau) \, d^n y \, d\tau.$$

Remark. The Green's function G_n is related to the fundamental solution S_n by $G_n = \Theta(t - \tau) S_n$. This reflects causality: the response at time t can only depend on sources at earlier times $\tau < t$.

Theorem (Duhamel's principle). The solution to the forced heat equation with homogeneous initial conditions can be written as

$$\phi(\mathbf{x}, t) = \int_0^t \phi_F(\mathbf{x}, t; \tau) \, d\tau,$$

where $\phi_F(\mathbf{x}, t; \tau)$ solves the *homogeneous* heat equation $\partial_t \phi_F = D \nabla^2 \phi_F$ for $t > \tau$, with initial condition $\phi_F(\mathbf{x}, \tau; \tau) = F(\mathbf{x}, \tau)$.

Remark. Duhamel's principle provides a physical interpretation: the forcing term $F(\mathbf{x}, t)$ can be viewed as imposing a sequence of “initial conditions” at each instant τ . The full solution is obtained by integrating the individual responses over all times $\tau \in [0, t]$.

Green's functions for the wave equation

We now consider the inhomogeneous wave equation

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \nabla^2 \phi = F(\mathbf{x}, t),$$

where $\phi : \mathbb{R}^n \times [0, \infty) \rightarrow \mathbb{R}$ and $c > 0$ is the wave speed. We impose homogeneous initial conditions:

$$\phi(\mathbf{x}, 0) = 0, \quad \partial_t \phi(\mathbf{x}, 0) = 0.$$

Definition (Green's function for the wave equation). The *Green's function* $G_n(\mathbf{x}, t; \mathbf{y}, \tau)$ for the wave equation in n spatial dimensions is the solution to

$$\frac{\partial^2 G_n}{\partial t^2} - c^2 \nabla^2 G_n = \delta(t - \tau) \delta^{(n)}(\mathbf{x} - \mathbf{y}),$$

with homogeneous initial conditions $G_n(\mathbf{x}, 0; \mathbf{y}, \tau) = \partial_t G_n(\mathbf{x}, 0; \mathbf{y}, \tau) = 0$.

Taking the Fourier transform with respect to the spatial variables $\mathbf{x}$:

$$\partial_t^2 \tilde{G}_n + c^2 |\mathbf{k}|^2 \tilde{G}_n = \delta(t - \tau) e^{-i\mathbf{k} \cdot \mathbf{y}}.$$

This is an ODE in t with solution

$$\tilde{G}_n(\mathbf{k}, t; \mathbf{y}, \tau) = \Theta(t - \tau) e^{-i\mathbf{k} \cdot \mathbf{y}} \frac{\sin |\mathbf{k}| c (t - \tau)}{|\mathbf{k}| c}.$$

The Green's function is recovered by inverse Fourier transform:

$$G_n(\mathbf{x}, t; \mathbf{y}, \tau) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} \Theta(t - \tau) \frac{\sin |\mathbf{k}| c (t - \tau)}{|\mathbf{k}| c} d^n k.$$

Remark. Unlike the heat equation, the form of the Green's function depends significantly on the number of spatial dimensions n . We compute the case $n = 3$ explicitly.

For $n = 3$, the Green's function is

$$G(\mathbf{x}, t; \mathbf{y}, \tau) = \frac{\Theta(t - \tau)}{(2\pi)^3 c} \int_{\mathbb{R}^3} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} \frac{\sin |\mathbf{k}| c (t - \tau)}{|\mathbf{k}|} d^3 k.$$

We use spherical polar coordinates with the z -axis in k -space aligned along the direction of $\mathbf{x} - \mathbf{y}$. Hence $\mathbf{k} \cdot (\mathbf{x} - \mathbf{y}) = kr \cos \theta$, where $r = |\mathbf{x} - \mathbf{y}|$ and $k = |\mathbf{k}|$.

Note that nothing in our integral depends on φ , so we can pull out a factor of 2π , and get

$$G(\mathbf{x}, t; \mathbf{y}, \tau) = \frac{\Theta(t - \tau)}{(2\pi)^2 c} \int_0^\infty \int_0^\pi e^{ikr \cos \theta} \frac{\sin kc(t - \tau)}{k} k^2 \sin \theta d\theta dk$$

The next integral to do is the θ integral, which is straightforward since it is an exact differential. Setting $\alpha = c(t - \tau)$, we get

$$\begin{aligned}
 &= \frac{\Theta(t - \tau)}{(2\pi)^2 c} \int_0^\infty \left[\frac{e^{ikr} - e^{-ikr}}{ikr} \right] \frac{\sin kc(t - \tau)}{k} k^2 dk \\
 &= \frac{\Theta(t - \tau)}{(2\pi)^2 icr} \left[\int_0^\infty e^{ikr} \sin k\alpha dk - \int_0^\infty e^{-ikr} \sin k\alpha dk \right] \\
 &= \frac{\Theta(t - \tau)}{2\pi icr} \left[\frac{1}{2\pi} \int_{-\infty}^\infty e^{ikr} \sin k\alpha dk \right] \\
 &= \frac{\Theta(t - \tau)}{2\pi icr} \mathcal{F}^{-1}[\sin k\alpha],
 \end{aligned}$$

Now recall $\mathcal{F}[\delta(x - \alpha)] = e^{-ik\alpha}$. So

$$\mathcal{F}^{-1}[\sin k\alpha] = \mathcal{F}^{-1} \left[\frac{e^{ik\alpha} - e^{-ik\alpha}}{2i} \right] = \frac{1}{2i} [\delta(x + \alpha) - \delta(x - \alpha)].$$

Since the first delta function $\delta(|\mathbf{x} - \mathbf{y}| + c(t - \tau))$ has a strictly positive argument when $t > \tau$, it does not contribute. The second delta function contributes only when $|\mathbf{x} - \mathbf{y}| = c(t - \tau)$.

Proposition (Green's function for the 3D wave equation). The Green's function for the wave equation in three spatial dimensions is

$$G_3(\mathbf{x}, t; \mathbf{y}, \tau) = \frac{1}{4\pi c |\mathbf{x} - \mathbf{y}|} \delta(|\mathbf{x} - \mathbf{y}| - c(t - \tau)).$$

The solution to the forced wave equation with homogeneous initial conditions is

$$\phi(\mathbf{x}, t) = \int_0^\infty \int_{\mathbb{R}^3} F(\mathbf{y}, \tau) G_3(\mathbf{x}, t; \mathbf{y}, \tau) d^3y d\tau.$$

Definition (Retarded time). The *retarded time* at position $\mathbf{x}$ due to a source at position $\mathbf{y}$ is

$$t_{\text{ret}} = t - \frac{|\mathbf{x} - \mathbf{y}|}{c}.$$

This is the time at which a signal must leave $\mathbf{y}$, travelling at speed c , in order to arrive at $\mathbf{x}$ at time t .

Using the delta function to perform the time integral, the solution becomes

$$\phi(\mathbf{x}, t) = \frac{1}{4\pi c^2} \int_{\mathbb{R}^3} \frac{F(\mathbf{y}, t_{\text{ret}})}{|\mathbf{x} - \mathbf{y}|} d^3y.$$

Remark (Finite propagation speed). The retarded time formula shows that the effect of the forcing term at $\mathbf{y}$ reaches the point $\mathbf{x}$ only after time $|\mathbf{x} - \mathbf{y}|/c$ has elapsed. This confirms that information propagates at

finite speed c , consistent with our analysis of characteristics for hyperbolic equations.

Remark (Geometric spreading). The factor $1/|\mathbf{x}-\mathbf{y}|$ represents *geometric spreading*: as distance increases, the effect of a source diminishes because its energy is distributed over a larger sphere. This dimension-dependent behaviour contrasts with the one-dimensional case, where the Green's function is

$$G_1(x, t; y, \tau) = \frac{1}{2c} \Theta(c(t - \tau) - |x - y|),$$

which has no decay factor. In one dimension, there is no spreading, so the signal amplitude is preserved.

7.5 Poisson's equation

We now study Poisson's equation in $\mathbb{R}^3$:

$$\nabla^2 \phi = -F,$$

where $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}$ is the unknown function and $F : \mathbb{R}^3 \rightarrow \mathbb{R}$ is a given source term.

Definition (Fundamental solution of Poisson's equation). The *fundamental solution* (or *free-space Green's function*) for the Laplacian in $\mathbb{R}^3$ is the function $G_3(\mathbf{x}, \mathbf{y})$ satisfying

$$\nabla_{\mathbf{x}}^2 G_3(\mathbf{x}, \mathbf{y}) = \delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

Proposition (Fundamental solution in 3D). The fundamental solution for the Laplacian in $\mathbb{R}^3$ is

$$G_3(\mathbf{x}, \mathbf{y}) = -\frac{1}{4\pi|\mathbf{x} - \mathbf{y}|},$$

where we choose the constant of integration so that $G_3 \rightarrow 0$ as $|\mathbf{x}| \rightarrow \infty$.

Proof. By rotational symmetry, G_3 depends only on $r = |\mathbf{x} - \mathbf{y}|$. Integrating over a ball B_r of radius r centred at $\mathbf{y}$ and using the divergence theorem:

$$1 = \int_{B_r} \nabla^2 G_3 \, dV = \int_{\partial B_r} \mathbf{n} \cdot \nabla G_3 \, dS = 4\pi r^2 \frac{dG_3}{dr}.$$

Hence $dG_3/dr = 1/(4\pi r^2)$, which integrates to $G_3 = -1/(4\pi r) + c$. Setting $c = 0$ ensures $G_3 \rightarrow 0$ as $r \rightarrow \infty$. $\square$

Green's identities

To make use of the fundamental solution in solving Poisson's equation, we first establish some useful identities. Let $\Omega \subseteq \mathbb{R}^3$ be a bounded region with smooth boundary $\partial\Omega$, and let $\mathbf{n}$ denote the outward unit normal to $\partial\Omega$.

Proposition (Green's first identity). For smooth functions $\phi, \psi : \bar{\Omega} \rightarrow \mathbb{R}$,

$$\int_{\partial\Omega} \phi \mathbf{n} \cdot \nabla \psi \, dS = \int_{\Omega} \phi \nabla^2 \psi + (\nabla \phi) \cdot (\nabla \psi) \, dV.$$

Proof. Apply the divergence theorem to $\nabla \cdot (\phi \nabla \psi) = \phi \nabla^2 \psi + (\nabla \phi) \cdot (\nabla \psi)$. $\square$

By symmetry, interchanging ϕ and ψ :

$$\int_{\partial\Omega} \psi \mathbf{n} \cdot \nabla \phi \, dS = \int_{\Omega} \psi \nabla^2 \phi + (\nabla \phi) \cdot (\nabla \psi) \, dV.$$

Subtracting these two equations yields:

Proposition (Green's second identity). For smooth functions $\phi, \psi : \bar{\Omega} \rightarrow \mathbb{R}$,

$$\int_{\Omega} (\phi \nabla^2 \psi - \psi \nabla^2 \phi) \, dV = \int_{\partial\Omega} (\phi \mathbf{n} \cdot \nabla \psi - \psi \mathbf{n} \cdot \nabla \phi) \, dS.$$

Remark. Green's second identity relates volume integrals involving $\nabla^2 \phi$ and $\nabla^2 \psi$ (which are determined by the PDEs) to boundary integrals of ϕ , ψ , and their normal derivatives (which are related to boundary conditions). This makes the identity particularly useful for solving boundary value problems.

Using the Green's function

We wish to apply this result to the case $\psi = G_3(|\mathbf{x} - \mathbf{y}|)$. However, recall that when deriving Green's identity, we assumed ϕ and ψ are smooth everywhere in our domain. However, our Green's function is singular at $\mathbf{x} = \mathbf{y}$, but we want to integrate over this region as well. So we need to do this carefully. Because of the singularity, we want to take

$$\Omega = B_r - B_\varepsilon = \{\mathbf{x} \in \mathbb{R}^3 : \varepsilon \leq |\mathbf{x} - \mathbf{y}| \leq R\}.$$

In other words, we remove a small region of radius ε centered on $\mathbf{y}$ from the domain.

In this choice of Ω , it is completely safe to use Green's identity, since our Green's function is certainly regular everywhere in this Ω . First note that since $\nabla^2 G_3 = 0$ everywhere except at $\mathbf{x} = \mathbf{y}$, we get

$$\int_{\Omega} \phi \nabla^2 G_3 - G_3 \nabla^2 \phi \, dV = - \int_{\Omega} G_3 \nabla^2 \phi \, dV$$

Then Green's second identity gives

$$\begin{aligned} - \int_{\Omega} G_3 \nabla^2 \phi \, dV &= \int_{S_r^2} \phi(\mathbf{n} \cdot \nabla G_3) - G_3(\mathbf{n} \cdot \phi) \, dS \\ &\quad + \int_{S_\varepsilon^2} \phi(\mathbf{n} \cdot \nabla G_3) - G_3(\mathbf{n} \cdot \phi) \, dS \end{aligned}$$

Note that on the inner boundary, we have $\mathbf{n} = -\hat{\mathbf{r}}$. Also, at S_ε^2 , we have

$$G_3|_{S_\varepsilon^2} = -\frac{1}{4\pi\varepsilon}, \quad \left. \frac{dG_3}{dr} \right|_{S_\varepsilon^2} = \frac{1}{4\pi\varepsilon^2}.$$

So the inner boundary terms are

$$\begin{aligned} & \int_{S_\varepsilon^2} \left(\phi(\mathbf{n} \cdot \nabla G_3) - G_3(\mathbf{n} \cdot \nabla \phi) \right) \varepsilon^2 \sin \theta \, d\theta \, d\phi \\ &= -\frac{\varepsilon^2}{4\pi\varepsilon^2} \int_{S_\varepsilon^2} \phi \sin \theta \, d\theta \, d\phi + \frac{\varepsilon^2}{4\pi\varepsilon} \int_{S_\varepsilon^2} (\mathbf{n} \cdot \nabla \phi) \sin \theta \, d\theta \, d\phi \end{aligned}$$

Now the final integral is bounded by the assumption that ϕ is everywhere smooth. So as we take the limit $\varepsilon \rightarrow 0$, the final term vanishes. In the first term, the ε 's cancel. So we are left with

$$\begin{aligned} &= -\frac{1}{4\pi} \int_{S_\varepsilon^2} \phi \, d\Omega \\ &= -\bar{\phi} \\ &\rightarrow -\phi(\mathbf{y}) \end{aligned}$$

where $\bar{\phi}$ is the average value of ϕ on the sphere.

Now suppose $\nabla^2 \phi = -F$. Then this gives

Proposition (Green's third identity).

$$\phi(\mathbf{y}) = \int_{\partial\Omega} \phi(\mathbf{n} \cdot \nabla G_3) - G_3(\mathbf{n} \cdot \nabla \phi) \, dS - \int_{\Omega} G_3(\mathbf{x}, \mathbf{y}) F(\mathbf{x}) \, d^3x.$$

Remark. Green's third identity expresses ϕ at any interior point $\mathbf{y}$ in terms of the fundamental solution G_3 , the source term F , and boundary data. In the special case where boundary contributions vanish as $|\mathbf{x}| \rightarrow \infty$, we obtain

$$\phi(\mathbf{y}) = - \int_{\mathbb{R}^3} G_3(\mathbf{x}, \mathbf{y}) F(\mathbf{x}) \, d^3x.$$

Thus the fundamental solution serves as the Green's function for Poisson's equation on all of $\mathbb{R}^3$.

Remark (The overspecification problem). Green's third identity requires knowledge of *both* ϕ and $\mathbf{n} \cdot \nabla \phi$ on the boundary. However, for Laplace's equation ($F = 0$), uniqueness theorems tell us that specifying either $\phi|_{\partial\Omega}$ (Dirichlet) or $\mathbf{n} \cdot \nabla \phi|_{\partial\Omega}$ (Neumann) alone determines the solution. Thus Green's third identity, while valid, is not directly constructive—we cannot specify both boundary conditions freely.

What we need is a representation formula that uses only one type of boundary data.

Dirichlet Green's function

Definition (Dirichlet Green's function). The *Dirichlet Green's function* for a domain Ω is a function $G(\mathbf{x}, \mathbf{y})$ of the form

$$G(\mathbf{x}, \mathbf{y}) = G_3(\mathbf{x}, \mathbf{y}) + H(\mathbf{x}, \mathbf{y}),$$

where H is harmonic in Ω (i.e., $\nabla_{\mathbf{x}}^2 H = 0$), regular throughout Ω , and chosen so that

$$G(\mathbf{x}, \mathbf{y})|_{\mathbf{x} \in \partial\Omega} = 0.$$

Since H is harmonic, G satisfies the same equation as G_3 :

$$\nabla_{\mathbf{x}}^2 G(\mathbf{x}, \mathbf{y}) = \delta^{(3)}(\mathbf{x} - \mathbf{y}).$$

Applying Green's third identity with G instead of G_3 , and using $G|_{\partial\Omega} = 0$:

Proposition (Representation formula for Dirichlet problem). If $\nabla^2 \phi = -F$ in Ω , then

$$\phi(\mathbf{y}) = \int_{\partial\Omega} \phi \mathbf{n} \cdot \nabla G \, dS - \int_{\Omega} F(\mathbf{x}) G(\mathbf{x}, \mathbf{y}) \, d^3x.$$

This expresses ϕ at interior points using only Dirichlet boundary data $\phi|_{\partial\Omega}$.

Remark. Similarly, for Neumann boundary conditions, one seeks a *Neumann Green's function* satisfying $\mathbf{n} \cdot \nabla G|_{\partial\Omega} = \text{const}$, which eliminates the $\phi|_{\partial\Omega}$ term.

In general, finding the harmonic correction H with

$$H|_{\partial\Omega} = -G_3|_{\partial\Omega} = \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|} \Big|_{\partial\Omega}$$

is a difficult problem. However, the *method of images* provides explicit solutions for domains with sufficient symmetry.

Definition (Method of images). The *method of images* constructs the Dirichlet (or Neumann) Green's function by placing fictitious “image sources” outside the domain Ω . These image sources are positioned so that their contribution, combined with the original source at $\mathbf{y}$, satisfies the desired boundary condition on $\partial\Omega$.

Example (Method of images for the half-space). Suppose

$$\Omega = \{(x, y, z) \in \mathbb{R}^3 : z \geq 0\}.$$

We wish to find a solution to $\nabla^2 \phi = -F$ in Ω with $\phi \rightarrow 0$ rapidly as $|\mathbf{x}| \rightarrow \infty$ with boundary condition $\phi(x, y, 0) = g(x, y)$.

The fundamental solution

$$G_3(\mathbf{x}, \mathbf{x}_0) = -\frac{1}{4\pi} \frac{1}{|\mathbf{x} - \mathbf{x}_0|}$$

obeys all the conditions we need except

$$G_3|_{z=0} = -\frac{1}{4\pi} \frac{1}{[(x-x_0)^2 + (y-y_0)^2 + z_0^2]^{1/2}} \neq 0.$$

However, let $\mathbf{x}_0^R$ be the point $(x_0, y_0, -z_0)$. This is the reflection of $\mathbf{x}_0$ in the boundary plane $z = 0$. Since the point $\mathbf{x}_0^R$ is outside our domain, $G_3(\mathbf{x}, \mathbf{x}_0^R)$ obeys

$$\nabla^2 G_3(\mathbf{x}, \mathbf{x}_0^R) = 0$$

for all $\mathbf{x} \in \Omega$, and also

$$G_3(\mathbf{x}, \mathbf{x}_0^R)|_{z=0} = G_3(\mathbf{x}, \mathbf{x}_0)|_{z=0}.$$

Hence we take

$$G(\mathbf{x}, \mathbf{x}_0) = G_3(\mathbf{x}, \mathbf{x}_0) - G_3(\mathbf{x}, \mathbf{x}_0^R).$$

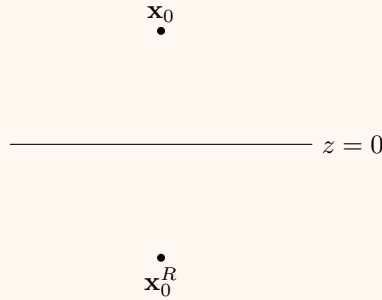
The outward pointing normal to Ω at $z = 0$ is $\mathbf{n} = -\hat{\mathbf{z}}$. Hence we have

$$\begin{aligned} \mathbf{n} \cdot \nabla G|_{z=0} &= \frac{1}{4\pi} \left[\frac{-(z-z_0)}{|\mathbf{x} - \mathbf{x}_0|^3} - \frac{-(z+z_0)}{|\mathbf{x} - \mathbf{x}_0^R|^3} \right]_{z=0} \\ &= \frac{1}{2\pi} \frac{z_0}{[(x-x_0)^2 + (y-y_0)^2 + z_0^2]^{3/2}}. \end{aligned}$$

Therefore our solution is

$$\begin{aligned} \phi(\mathbf{x}_0) &= \frac{1}{4\pi} \int_{\Omega} \left[\frac{1}{|\mathbf{x} - \mathbf{x}_0|} - \frac{1}{|\mathbf{x} - \mathbf{x}_0^R|} \right] F(\mathbf{x}) \, d^3x \\ &\quad + \frac{z_0}{2\pi} \int_{\mathbb{R}^2} \frac{g(x, y)}{[(x-x_0)^2 + (y-y_0)^2 + z_0^2]^{3/2}} \, dx \, dy. \end{aligned}$$

What have we actually done here? The Green's function $G_3(\mathbf{x}, \mathbf{x}_0)$ in some sense represents a “charge” at $\mathbf{x}_0$. We can imagine that the term $G_3(\mathbf{x}, \mathbf{x}_0^R)$ represents the contribution to our solution from a point charge of opposite sign located at $\mathbf{x}_0^R$. Then by symmetry, G is zero at $z = 0$.



Our solution for $\phi(\mathbf{x}_0)$ is valid if we extend it to $\mathbf{x}_0 \in \mathbb{R}^3$ where $\phi(\mathbf{x}_0)$ is solved by choosing two mirror charge distributions. But this is irrelevant for our present purposes. We are only concerned with finding a solution in the region $z \geq 0$. This is just an artificial device we have in order to solve the problem.

Example (Method of images for the heat equation). Suppose a chimney produces smoke such that the density $\phi(\mathbf{x}, t)$ of smoke obeys

$$\partial_t \phi - D \nabla^2 \phi = F(\mathbf{x}, t).$$

The left side is just the heat equation, modelling the diffusion of smoke, while the right forcing term describes the production of smoke by the chimney.

If this were a problem for $\mathbf{x} \in \mathbb{R}^3$, then the solution is

$$\phi(\mathbf{x}, t) = \int_0^t \int_{\mathbb{R}^3} F(\mathbf{y}, \tau) S_3(\mathbf{x} - \mathbf{y}, t - \tau) d^3 \mathbf{y} d\tau,$$

where

$$S_3(\mathbf{x} - \mathbf{y}, t - \tau) = \frac{1}{[4\pi D(t - \tau)]^{3/2}} \exp\left(-\frac{|\mathbf{x} - \mathbf{y}|^2}{4D(t - \tau)}\right).$$

This is true only if the smoke can diffuse in all of $\mathbb{R}^3$. However, this is not true for our current circumstances, since smoke does not diffuse into the ground.

To account for this, we should find a Green's function that obeys

$$\mathbf{n} \cdot \nabla G|_{z=0} = 0.$$

This says that there is no smoke diffusing in to the ground.

This is achieved by picking

$$G(\mathbf{x}, t; \mathbf{y}, \tau) = \Theta(t - \tau)[S_3(\mathbf{x} - \mathbf{y}, t - \tau) + S_3(\mathbf{x} - \mathbf{y}^R, t - \tau)].$$

We can directly check that this obeys

$$\partial_t D^2 \nabla^2 G = \delta(t - \tau) \delta^3(\mathbf{x} - \mathbf{y})$$

when $\mathbf{x} \in \Omega$, and also

$$\mathbf{n} \cdot \nabla G|_{z_0} = 0.$$

Hence the smoke density is given by

$$\phi(\mathbf{x}, t) = \int_0^t \int_{\Omega} F(\mathbf{y}, \tau)[S_3(\mathbf{x} - \mathbf{y}, t - \tau) + S_3(\mathbf{x} - \mathbf{y}^R, t - \tau)] d^3 y.$$

We can think of the second term as the contribution from a “mirror chimney”. Without a mirror chimney, we will have smoke flowing into the ground. With a mirror chimney, we will have equal amounts of mirror smoke flowing up from the ground, so there is no net flow. Of course, there are no mirror chimneys in reality. These are just artifacts we use to find the solution we want.

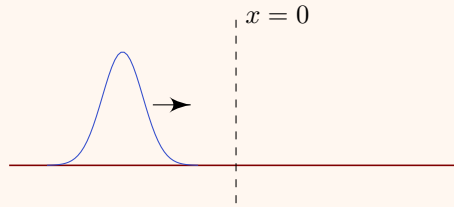
Example (Method of images for the wave equation). Suppose we want to solve the wave equation in the region (x, t) such that $x > 0$ with boundary conditions

$$\phi(x, 0) = b(x), \quad \partial_t \phi(x, 0) = 0, \quad \partial_x \phi(0, t) = 0.$$

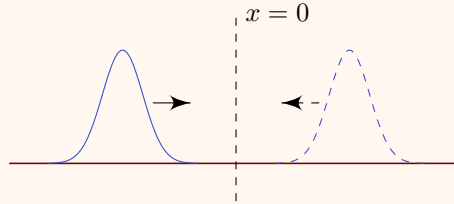
On $\mathbb{R}^{1,1}$ d'Alembert's solution gives

$$\phi(x, t) = \frac{1}{2}[b(x - ct) + b(x + ct)]$$

This is not what we want, since eventually we will have a wave moving past the $x = 0$ line.



To compensate for this, we introduce a mirror wave moving in the opposite direction, such that when as they pass through each other at $x = 0$, there is no net flow across the boundary.



More precisely, we include a *mirror initial condition* $\phi(x, 0) = b(x) + b(-x)$, where we set $b(x) = 0$ when $x < 0$. In the region $x > 0$ we are interested in, only the $b(x)$ term will contribute. In the $x < 0$ region, only $x > 0$ will contribute. Then the general solution is

$$\phi(x, t) = \frac{1}{2}[b(x + ct) + b(x - ct) + b(-x - ct) + b(-x + ct)].$$